Arithmetic Expression Geometry I: Foundations

Sequential Histories, Affine Flow, Torsion, and Contact Geometry

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Abstract

Arithmetic Expression Geometry (AEG) retains the ordered history of an arithmetic evaluation instead of identifying an expression with its endpoint value. We first classify the finite binary expression trees having a unique legal order of internal evaluation: their internal vertices form a single spine. Marking an accumulator on such a tree produces a chronological word of one-hole contexts, while keeping operand-slot chirality distinct from temporal reversal and path inverse. Non-degenerate bilateral contexts act projectively on $\mathbb{P}^1(K)$. Over an arbitrary field K, translations, nonzero scalings, and inversion generate $\mathrm{PGL}_2(K)$; the contexts fixing infinity form its affine Borel sector. The $\sqrt{2}$-translation and inversion sublanguage realizes the $q = 4$ Hecke group while preserving the history–operator–coset distinction. The positive real affine component carries two translation cocycles and yields

$$\frac{da}{ds} = \mu \cos\theta + \lambda a \sin\theta.$$

We define regular arithmetic expression spaces intrinsically and derive their arithmetic frame. For $\mu, \lambda > 0$ we construct a complete upper-half-plane model of curvature $-\lambda^2$ with assignment $a = -x/y$. When $\mu \neq 0$, zero is a regular value; if the AES is also connected, complete, and boundaryless, then $M \cong Z(a) \times \mathbb{R}$ and $Z(a)$ is connected. Isolated or branching zeros therefore require singular data or a failure of these global hypotheses. For positive add–scale histories, a direct cocycle gives charge-compatible pairs a torsion invariant and an exact weighted Stokes identity in the Accumulative Commutative Space. Finally, the contact form $\alpha = da - \mu\, du - \lambda a\, dv$ localizes the same order defect through $[D_u, D_v] = \mu\lambda\partial_a$. Exact open and closed finite defects are separated from their common infinitesimal curvature. These results establish the affine foundational sector of AEG inside its projective semantics; analytic function theory, singular tube topology, projective condensation, and complexity are left to later papers.

Keywords: arithmetic expressions; evaluation histories; projective linear group; affine group; Hecke group; hyperbolic geometry; cocycles; contact geometry; regular zero sets

Contents

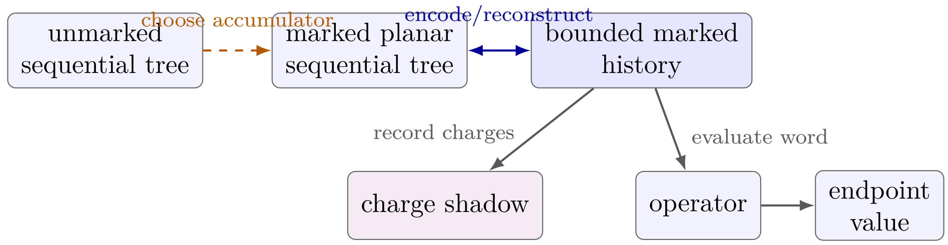

Figure 1: The process-to-result structure. Choosing an accumulator marks an unmarked sequential tree; a fully marked planar tree and a bounded marked history determine one another. From the history, operator evaluation and the charge map are separate information-losing branches, while an endpoint is obtained by applying the operator to an initial value.

1 Introduction

An arithmetic expression has several mathematically distinct lives. It is a finite syntax tree; it may be equipped with a legal order of evaluation; a chosen running value turns that order into a marked history; the history induces an operator; and the operator sends an initial value to an endpoint. Ordinary arithmetic usually passes through these levels as quickly as possible. AEG instead asks what is lost at each passage.

The guiding structure of this paper is

$$\begin{gathered}\text{unmarked sequential tree} \xrightarrow{\text{choose an accumulator}} \text{marked planar sequential tree},\\ \text{marked planar sequential tree} \longleftrightarrow \text{bounded marked history},\\ \text{bounded marked history} \longrightarrow \text{operator} \longrightarrow \text{endpoint value},\\ \text{bounded marked history} \longrightarrow \text{charge shadow}.\end{gathered} \tag{1}$$

The first arrow denotes a choice, not a canonical map. When the planar structure, mark, operation labels, constants, and slots are retained, the double arrow is the bijection proved in Section 2. The two arrows leaving a history are separate forgetful maps: charge is not a function of the evaluated operator, and an endpoint is not a function of charge alone. Different histories may nevertheless induce the same operator, and different operators may agree at one chosen initial value. The geometry below belongs primarily to these intermediate levels, before endpoint evaluation has erased the order of construction.

Figure 1 renders this structure explicitly. It is a map of the distinctions used in the paper, not a claim that any forgetful arrow is injective.

1.1 Sequential histories

Not every expression tree determines a unique history. Independent subexpressions may be evaluated in either order, so a single tree can have many legal linear extensions of its dependency order. Section 2 isolates the precise exception. For a finite binary tree, uniqueness of the internal evaluation order is equivalent to the internal vertices forming one chain, or equivalently one spine. A marked accumulator on that spine determines a word of one-hole contexts

$$C^{(1)}_{\omega,c}[z] = \omega(z,c), \qquad C^{(2)}_{\omega,c}[z] = \omega(c,z).$$

The superscript records the operand slot occupied by the current value. It is not a time direction. This distinction prevents three operations that are often blurred in informal tree pictures—planar mirror, temporal reversal, and path inverse—from being identified.

This use of syntax is modest. Formal generation and rewriting have long histories in logic and theoretical computer science; classical points of reference include Post's work on combinatorial reduction and Chomsky's generative hierarchies [5, 1]. Here the purpose is different: the sequential fragment supplies a controlled domain on which evaluation can be treated as a path without pretending that every arithmetic tree has a unique path.

1.2 Why bilateral completion matters

Earlier affine calculations use the current value in one distinguished operand slot. That fragment is closed under translations and nonzero scalings and is therefore naturally represented by the affine group of the line. Once both operand slots are allowed, the context $z \mapsto c/z$ introduces inversion. The correct algebraic ambient space is then the projective line and its fractional linear transformations.

Section 3 proves, over an arbitrary field K, that the non-degenerate bilateral contexts generate $\mathrm{PGL}_2(K)$. The affine transformations are exactly the stabilizer of the projective point at infinity. Thus the continuous geometry developed in the rest of this paper is neither discarded nor promoted to the whole theory: it is identified precisely as the positive real affine component of a larger projective semantics.

Inside that larger semantics, the two contexts

$$T(z) = z + \sqrt{2}, \qquad J(z) = -\frac{1}{z}$$

generate the $q = 4$ Hecke group. Section 3 verifies the relation $(JT)^4 = 1$ directly from the context matrices and formulates the resulting interface from arithmetic words to group operators and then to cell cosets. This restricted alphabet is important because its hyperbolic cellulations are produced by arithmetic operator semantics rather than imposed after endpoint evaluation. The interface is deliberately not a bijection: relations identify different histories, and stabilizers identify different operators at the level of cells or endpoints.

Projective continuation does not repair an inadmissible ordinary computation. For example, c/z has a well-defined projective value at $z = 0$, namely infinity, while ordinary division by zero remains undefined. We keep the ordinary domain and the projective extension as separate data throughout.

1.3 Main results

The first group of results is algebraic. The full bilateral language maps onto $\mathrm{PGL}_2(K)$, while the distinguished $\mathbb{Q}(\sqrt{2})$-sublanguage above records an order-four relation inside a restricted projective subgroup. With a fixed chronological convention, an affine history

$$f_i(x) = s_i x + t_i, \qquad s_i \neq 0,$$

has total scale and translation

$$\Phi_n = \prod_{i=1}^{n} s_i, \qquad \xi_n = \sum_{i=1}^{n} t_i \prod_{j=i+1}^{n} s_j.$$

The normalized coordinate $\widehat{\xi}_n = \xi_n/\Phi_n$ gives a second, source-frame cocycle. These two coordinates make explicit which portion of a past translation is rescaled by future multiplication. Equal total scale is exactly the condition under which the endpoint difference of two affine histories is independent of the initial value.

The second group of results passes from finite histories to continuous propagation. After choosing additive and multiplicative intensities μ and λ, the affine Lie algebra yields

$$\frac{da}{ds} = \mu \cos\theta + \lambda a \sin\theta. \tag{2}$$

The angle is measured in an oriented orthonormal arithmetic frame, not in an arbitrary coordinate chart. We give an intrinsic metric definition of a regular arithmetic expression space and prove its equivalence, when $\mu \neq 0$, to the framed directional formulation. The eikonal identity

$$|\nabla a|_g^2 = \mu^2 + \lambda^2 a^2$$

is consequently metric-dependent rather than a coordinate-free statement without geometric data.

The basic homogeneous realization is the upper half-plane with

$$g_{\mu,\lambda} = \frac{1}{y^2}\left(\frac{dx^2}{\mu^2} + \frac{dy^2}{\lambda^2}\right), \qquad a(x,y) = -\frac{x}{y}, \qquad \mu, \lambda > 0.$$

We derive this metric from a normalized left-invariant frame on $\mathrm{Aff}^+(1,\mathbb{R})$, verify the eikonal equation, and compute its curvature and Laplacian. This is a basic homogeneous model after normalization, not a uniqueness theorem.

The eikonal identity has an immediate topological consequence that was absent from earlier versions of the theory. At a zero of a one has $|\nabla a| = |\mu|$. When $\mu \neq 0$, zero is therefore a regular value, and the interior zero locus on a surface is a disjoint union of smooth curves. Isolated zeros, crossings, and branch points cannot occur in the regular category. We also prove a global strengthening: on a connected, complete, boundaryless regular AES, rectifying the assignment produces a complete unit-gradient flow and a diffeomorphism

$$M \cong Z(a) \times \mathbb{R}.$$

Thus the zero set is nonempty and connected; the statement is a smooth splitting, not necessarily a Riemannian product. We accordingly define singular arithmetic expression spaces by a closed nowhere-dense singular set with no local regular extension, and reclassify the radial isolated-zero example by its failure of differentiability at the center.

The third group of results compares order globally and locally. The Accumulative Commutative Space (ACS) records the total additive charge A and logarithmic scale charge M of a positive add–scale history. Its terminal charge forgets order, but the ordered path traced in the (A, M)-plane retains enough information to reconstruct evaluation:

$$\nu_x(\gamma) = e^{M_\gamma}\left(x + \int_{C_\gamma} e^{-M}\, dA\right).$$

For a charge-compatible pair of such histories, their relative affine defect is a weighted boundary integral and, by Stokes' theorem, a weighted area integral. This is the exact global meaning of arithmetic torsion used here. The term denotes an order defect; it is not the torsion tensor of an affine connection.

Finally, when $\mu\lambda \neq 0$, the three-dimensional state space with contact form

$$\alpha = da - \mu\, du - \lambda a\, dv$$

provides the local history model. Its preferred horizontal lifts satisfy

$$D_u = \partial_u + \mu \partial_a, \qquad D_v = \partial_v + \lambda a \partial_a, \qquad [D_u, D_v] = \mu\lambda\partial_a.$$

We distinguish a target-frame open two-path endpoint defect, a source-normalized closed commutator holonomy, and the common infinitesimal curvature density. The associated horizontal covariant differential on scalar fields is curvature-sensitive and is not a nilpotent de Rham complex.

1.4 Scope and relation to standard geometry

Several ingredients used here are standard: fractional linear transformations, semidirect-product cocycles, left-invariant metrics, the regular-value theorem, Stokes' theorem, and contact structures. The mathematical contribution sought in Paper I is their history-sensitive arithmetic organization and the exact interfaces among them. In particular, the contact form is locally a standard contact form; its distinctive feature here is the interpretation of its chosen horizontal frame, not a new contact-isomorphism class. Standard references for the differential-geometric and contact background include [3, 2].

Four boundaries are enforced. First, a horizontal contact distribution does not by itself select a unique complex structure, so arithmetic holomorphic and harmonic theories are deferred to Paper II; this includes any automorphic-function theory built on the G_4 action. Second, the regular-value theorem is only the starting point for singular families: pulling analytic real loci back to the Hecke cellulation, and developing their multiple zeros, discriminants, tubes, braids, and knots, belong to Paper III. Third, projective bivaluations, homogeneous quotient towers, and condensation belong to Paper IV. Fourth, no computational lower bound follows from noncommutativity or negative curvature alone; complexity claims require an explicit state space, encoding, metric, and cost model.

1.5 Organization

Section 2 develops sequential trees and marked histories. Section 3 gives the bilateral projective semantics and locates the affine sector. Section 4 proves the affine cocycle formulas and defines relative affine defects. Section 5 derives the continuous flow and fixes the regular-AES definition. Section 6 constructs the homogeneous hyperbolic model, and Section 7 develops its regular and singular zero geometry. Section 8 proves the global weighted Stokes identity. Section 9 gives the contact localization and the exact finite-to-infinitesimal comparison. Section 10 records the proved interface exported to the later papers. The appendices collect convention checks, cocycle derivations, model calculations, and small examples separating equality levels.

2 Sequential Arithmetic Histories

Arithmetic evaluation is usually presented as a map from a syntactic expression to its value. This description suppresses the order in which intermediate values are formed. We begin by isolating the finite binary trees for which the internal operations have only one legal temporal order. Such a tree is not yet a history: one must also mark which leaf of its innermost operation is to be carried forward as the evolving value. The resulting object, a marked spinal history, is the syntactic unit used in the remainder of the paper.

Throughout this section, K is a field and

$$\Omega = \{+, -, \times, /\}$$

is the binary arithmetic signature. Division is interpreted as a partial operation in ordinary arithmetic. The combinatorial statements below apply to any binary signature; the choice of Ω is needed only when histories are evaluated.

2.1 Expression trees and dependency

Definition 2.1 (Arithmetic expression tree)**.** An *arithmetic expression tree over K* is a finite rooted planar binary tree whose leaves are labelled by elements of K or by formal variables and whose internal vertices are labelled by elements of Ω. Its set of internal vertices is denoted by $I(T)$.

The planar structure records the first and second operand slots. It does not, by itself, specify a temporal order between operations on disjoint subtrees. That order is recorded by the dependency relation.

Definition 2.2 (Internal dependency poset)**.** For $u, v \in I(T)$, write $u \preccurlyeq_T v$ when either $u = v$ or the value produced at u is required as an input at v. Equivalently, v lies on the path from u to the root. A *legal internal evaluation order* is a linear extension of the finite poset $(I(T), \preccurlyeq_T)$.

Leaves are not included in $I(T)$. Thus a tree with no internal vertex has the unique empty internal evaluation order. This convention separates the order of operations from the prior availability of atomic inputs.

We shall use the following elementary finite-poset fact in both directions.

Lemma 2.3 (Unique linear extensions)**.** *A finite poset has a unique linear extension if and only if it is a chain.*

Proof. Every linear extension of a chain is its given total order, so a chain has a unique linear extension. Conversely, let P be a finite poset and suppose that $u, v \in P$ are incomparable. Adding the relation $u < v$ and taking its transitive closure cannot create a directed cycle: such a cycle would contain a pre-existing path $v \leq u$, contrary to incomparability. The resulting finite partial order has a linear extension. The same argument with $v < u$ gives a second linear extension. One orders u before v and the other orders v before u, so the extension of P is not unique. Hence a poset with a unique linear extension contains no incomparable pair and is a chain. □

Theorem 2.4 (Sequential-tree classification)**.** *For a finite binary expression tree T, the following conditions are equivalent.*

1. *The dependency poset $I(T)$ has a unique linear extension.*
2. *The dependency poset $I(T)$ is a chain.*
3. *Every internal vertex has at most one internal child.*
4. *The internal vertices form a single spine: either $I(T)$ is empty, or there is a path from the root to an innermost internal vertex containing every internal vertex of T.*

Proof. The equivalence of the first two conditions is Lemma 2.3.

Suppose that $I(T)$ is a chain. If an internal vertex had two internal children, neither child could depend on the other, since they lie in disjoint subtrees. They would therefore be incomparable in $I(T)$, a contradiction. Thus (2) implies (3).

Assume (3). If $I(T)$ is empty there is nothing to prove. Otherwise begin at the root. Whenever the current internal vertex has an internal child, that child is unique. Following these children produces a path ending at an internal vertex with two leaf children. Every internal vertex lies on this path: an internal vertex off the path would occur in the other child-subtree of some vertex on the path, giving that vertex two internal children. Hence the path contains all internal vertices, proving (4).

Finally, if all internal vertices lie on one root-to-leaf path, their ancestor relation is total. Read from the innermost vertex towards the root, each operation depends on its predecessor. Thus $I(T)$ is a chain, proving (4)⇒(2) and completing the equivalence. □

Figure 2 contrasts the two possible local patterns at the level of internal dependencies. It is only a visual summary of the order relation; the classification rests on the preceding proof.

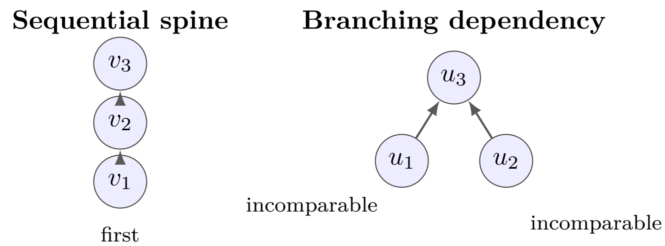

Figure 2: A sequential tree has a chain of internal dependencies (left). A branching tree contains incomparable internal operations and therefore admits more than one legal internal evaluation order (right).

Definition 2.5 (Sequential tree)**.** A finite binary expression tree satisfying the equivalent conditions of Theorem 2.4 is called a *sequential tree*.

The adjective refers only to internal evaluation. It does not choose an active leaf, and it does not assert that ordinary arithmetic evaluation is defined. For example,

$$(a+b)(c+d)$$

is not sequential: the two additions are incomparable dependencies of the final multiplication. By contrast,

$$((a+b)c-d)/e$$

is sequential, because its internal vertices lie on one spine.

2.2 One-hole contexts and the marked accumulator

Let z denote a hole rather than an arithmetic variable. For $\omega \in \Omega$ and $c \in K$, define the two elementary one-hole contexts

$$C^{(1)}_{\omega,c}[z] = \omega(z,c), \qquad C^{(2)}_{\omega,c}[z] = \omega(c,z). \tag{3}$$

The superscript gives the operand slot occupied by the evolving value. It is therefore intrinsic to the marked process and should not be replaced by a description based on how a planar drawing happens to expand.

At the innermost internal vertex of a nontrivial sequential tree, both children are leaves. The dependency poset determines that this operation is performed first, but it does not determine which of its two inputs is the value to be carried along the spine. Choosing one of those leaves as the initial accumulator removes this ambiguity.

Definition 2.6 (Marked spinal history)**.** A *free marked spinal history over K* is a finite chronological word

$$\gamma = (g_1, \ldots, g_n), \qquad g_i = C^{(\varepsilon_i)}_{\omega_i,c_i}, \quad \varepsilon_i \in \{1,2\}.$$

The context g_1 is applied first and g_n last. Its *chirality word* is

$$\varepsilon(\gamma) = \varepsilon_1 \cdots \varepsilon_n.$$

A *bounded marked spinal history* is a pair $(x_0;\gamma)$ consisting of an initial accumulator x_0 and a free history. The empty word is allowed; its bounded form represents the marked single-leaf tree.

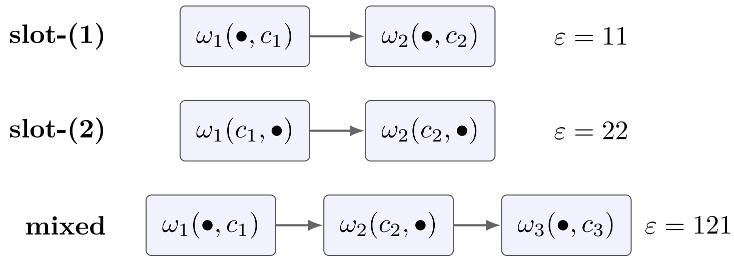

Figure 3: Pure slot-(1), pure slot-(2), and mixed marked histories. In each box, $\bullet$ is the current accumulator—the marked seed in the first box and the preceding output thereafter. Its displayed position is the operand slot, while time always runs along the arrows.

When no confusion is possible, we call either form a marked spinal history and state explicitly whether an initial value is present. Earlier drafts called certain pure-chirality cases threadlike expressions. The present definition also includes mixed chirality and makes the marking part of the data.

Figure 3 displays operand-slot chirality directly in the one-hole contexts. The diagram is a notation guide, not a substitute for the marked-tree correspondence proved below.

Proposition 2.7 (Marked-tree–history correspondence)**.** *Let T be a sequential tree all of whose leaves are labelled by elements of K.*

1. *If T is a single leaf, marking that leaf determines the empty bounded history.*
2. *If T is nontrivial, a choice of one leaf at its innermost internal vertex as the initial accumulator determines a unique bounded marked spinal history.*
3. *Conversely, a bounded marked spinal history determines a unique marked planar sequential tree.*

These two constructions are mutually inverse when operation labels, constants, operand slots, and the initial mark are retained.

Proof. The single-leaf case is immediate. Suppose T is nontrivial. By Theorem 2.4, its internal vertices form a unique spine. At the innermost vertex, mark one leaf x_0 and denote the other by c_1. The operation label and the slot containing x_0 determine $g_1 = C^{(\varepsilon_1)}_{\omega_1,c_1}$. Move one vertex towards the root. Its off-spine child must be a leaf, since otherwise it would be a second internal child. Its label c_2, the operation label, and the slot occupied by the spine determine g_2. Repeating this procedure reaches the root and produces a unique chronological word $(g_1, \ldots, g_n)$.

Conversely, start with a marked leaf x_0. Replace it by the planar binary tree $C^{(\varepsilon_1)}_{\omega_1,c_1}[x_0]$, then place the resulting tree in slot ε_2 of the next context, and continue. Exactly one child at every new internal vertex is internal, so the result is sequential. Each step records the operation, off-spine leaf, and slot used in the forward construction. The constructions therefore recover one another. $\square$

The proposition concerns fully labelled planar trees. If parentheses, constant labels, or the marked seed are forgotten, the correspondence is no longer injective.

2.3 Composition and ordinary evaluation

If

$$\gamma = (g_1, \ldots, g_m), \qquad \delta = (h_1, \ldots, h_n),$$

their chronological concatenation is

$$\gamma\delta = (g_1, \dots, g_m, h_1, \dots, h_n), \tag{4}$$

so γ is performed first and δ second. This convention is fixed for the entire paper and is summarized again in Appendix A.

Each elementary context determines a partial function $g_i \colon K \dashrightarrow K$. Starting from $x \in K$, define recursively

$$x_0 = x, \qquad x_i = g_i(x_{i-1}).$$

The bounded history is *ordinarily admissible at* x when every x_i is defined in ordinary field arithmetic. In that case its endpoint is

$$\nu_x(\gamma) = g_n \circ \cdots \circ g_1(x). \tag{5}$$

The admissible domain of a free history is the set of such x. In particular, a second-slot division c/z excludes any state at which the evolving value is zero, while a first-slot division z/c requires $c \neq 0$.

The concatenation convention gives

$$\nu_x(\gamma\delta) = \nu_{\nu_x(\gamma)}(\delta) \tag{6}$$

whenever both sides are ordinarily admissible. Equation (6) is a statement about partial functions. Chapter 3 introduces a projective evaluation with a different domain convention.

2.4 Mirror, temporal reversal, and path inverse

Three operations on a context word must be kept separate.

Definition 2.8 (Mirror and temporal reversal)**.** For $g = C^{(\varepsilon)}_{\omega,c}$, let

$$m(g) = C^{(3-\varepsilon)}_{\omega,c}.$$

The *mirror* and *temporal reversal* of $\gamma = (g_1, \dots, g_n)$ are respectively

$$m\gamma = (m(g_1), \dots, m(g_n)), \qquad r\gamma = (g_n, \dots, g_1).$$

The mirror exchanges operand slots while retaining chronological order; temporal reversal reverses chronological order while retaining the contexts.

Both operations are involutions on marked words, and they commute: $m(r\gamma) = r(m\gamma)$. This is a syntactic statement; ordinary domains of the evaluated words need not agree.

Definition 2.9 (Path inverse)**.** Suppose every context g_i in γ acts as an invertible function on a declared domain. The *path inverse* is the functional word

$$\gamma^{-1} = (g_n^{-1}, \dots, g_1^{-1}).$$

It is defined only under this invertibility hypothesis and need not have the same operation labels or ordinary admissible domain as $r\gamma$.

For example, let $g_1[z] = z + 1$ and $g_2[z] = 2z$. Then

$$\gamma = (g_1, g_2) \colon x \longmapsto 2x + 2.$$

Because addition and multiplication are symmetric operations, $m\gamma$ has the displayed expression $2(1+x)$ and induces the same operator $2x+2$. In contrast,

$$r\gamma = (g_2, g_1)\colon x \longmapsto 2x+1,$$

while

$$\gamma^{-1} = (z \mapsto z/2,\ z \mapsto z-1)\colon x \longmapsto x/2-1.$$

Thus equality with a mirror in this example is caused by the symmetry of the chosen elementary operations; it does not identify mirror, reversal, and inverse. The affine order defect studied later compares temporal orders. It is not automatically an invariant of planar mirror.

2.5 Information retained and information forgotten

Several equivalence relations occur naturally, and the remainder of the paper uses them at different stages. It is useful to order them by the information they forget.

1. *Literal syntactic equality* retains the fully parenthesized written expression.
2. *Marked-history equality* retains the initial mark, the chronological context word, and every operand-slot label. It may forget inessential typography but not process data.
3. *Operator equality* identifies histories inducing the same partial function on a stated ordinary domain, or the same projective transformation after projective evaluation.
4. *Charge equality*, when a charge map has been specified, retains only the chosen abelian totals of a history.
5. *Endpoint equality* at x asks only whether the admissible values $\nu_x(\gamma)$ and $\nu_x(\delta)$ coincide.
6. *Quotient or relational equality* is any further identification imposed by an explicitly declared rewriting or condensation relation.

This list is not a claim that every adjacent pair gives a strict quotient in all languages. It is a discipline for stating which level is being used. For example, the two histories

$$\gamma = (z \mapsto z+1,\ z \mapsto 2z), \qquad \delta = (z \mapsto 3z,\ z \mapsto z+1)$$

are distinct marked histories and induce the distinct operators $2z+2$ and $3z+1$, but at the initial value $x=1$ both have endpoint 4.

2.6 Representative chirality patterns

The slot notation exposes examples that the older comb terminology obscured.

Example 2.10 (Pure slot-(1) history)**.** The word

$$C^{(1)}_{+,2}, \quad C^{(1)}_{\times,3}, \quad C^{(1)}_{-,4}$$

has chirality word 111 and evaluates ordinarily as $x \mapsto (x+2)3-4$.

Example 2.11 (Pure slot-(2) history)**.** The word

$$C^{(2)}_{-,2}, \quad C^{(2)}_{/,3}$$

has chirality word 22 and evaluates as

$$x \longmapsto \frac{3}{2-x},$$

on the ordinary domain $K \setminus \{2\}$. The accumulator occupies the second operand slot at both steps.

Example 2.12 (Mixed chirality)**.** The word

$$C^{(1)}_{-,1}, \quad C^{(2)}_{/,2}, \quad C^{(1)}_{+,3}$$

has chirality word 121 and evaluates as

$$x \longmapsto \frac{2}{x-1} + 3,$$

on its ordinary domain. Its dependency poset is still a chain; mixed chirality does not introduce branching.

The syntactic passage is therefore

$$\begin{gathered} \text{unmarked sequential tree} \xrightarrow{\text{choice of accumulator}} \text{marked planar sequential tree}, \\ \text{marked planar sequential tree} \longleftrightarrow \text{bounded marked spinal history}, \\ \text{bounded marked spinal history} \longrightarrow \text{operator} \longrightarrow \text{endpoint}. \end{gathered}$$

when the history is bounded and all planar labels are retained. The first arrow is a choice and the double arrow is a bijection; only the later evaluation arrows forget process data. Chapter 3 constructs the operator stage for all non-degenerate bilateral contexts.

3 Projective Semantics and the Affine Sector

The four arithmetic operations behave asymmetrically once the accumulator may occupy either operand slot. Most elementary contexts are affine, but the second-slot division c/z is genuinely projective. This section makes that enlargement precise. The projective semantics does not replace ordinary arithmetic evaluation: it supplies an operator-valued continuation on a different domain.

Throughout this section, K is an arbitrary field. Points of the projective line are one-dimensional subspaces of K^2, written in homogeneous coordinates as

$$[u : v], \qquad (u, v) \neq (0, 0).$$

We use the affine chart $z = [z : 1]$ and write $\infty = [1 : 0]$. Matrices act on column vectors from the left. Thus

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \quad \Longrightarrow \quad M \cdot z = \frac{Az + B}{Cz + D}, \tag{7}$$

with the homogeneous interpretation at poles and at infinity. Replacing M by a nonzero scalar multiple does not change the projective map.

3.1 Ordinary domains and projective continuation

For an elementary context $g = C^{(\varepsilon)}_{\omega,c}$, ordinary evaluation gives a partial function $g\colon K \dashrightarrow K$. A *non-degenerate projective context* is one for which the corresponding homogeneous matrix is invertible. It acts everywhere on $\mathbb{P}^1(K)$, although its affine-coordinate formula may have a pole.

The elementary contexts and matrix representatives are as follows.

Context	Affine-coordinate map	Matrix representative
$C^{(1)}_{+,c} = C^{(2)}_{+,c}$	$z \mapsto z + c$	$\begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix}$
$C^{(1)}_{-,c}$	$z \mapsto z - c$	$\begin{pmatrix} 1 & -c \\ 0 & 1 \end{pmatrix}$
$C^{(2)}_{-,c}$	$z \mapsto c - z$	$\begin{pmatrix} -1 & c \\ 0 & 1 \end{pmatrix}$
$C^{(1)}_{\times,c} = C^{(2)}_{\times,c}$	$z \mapsto cz$	$\begin{pmatrix} c & 0 \\ 0 & 1 \end{pmatrix}$
$C^{(1)}_{/,c}$	$z \mapsto z/c$	$\begin{pmatrix} 1 & 0 \\ 0 & c \end{pmatrix}$
$C^{(2)}_{/,c}$	$z \mapsto c/z$	$\begin{pmatrix} 0 & c \\ 1 & 0 \end{pmatrix}$

The first three rows are invertible for every $c \in K$. Multiplication by c, first-slot division by c, and second-slot division with numerator c are projectively invertible precisely when $c \neq 0$. At $c = 0$, multiplication and $0/z$ are constant on their ordinary domains, while $z/0$ has empty ordinary domain. Their singular matrix representatives do not act everywhere on $\mathbb{P}^1(K)$: each has a projective base point. They are therefore excluded from $\mathrm{PGL}_2(K)$ and from the non-degenerate history language used below.

The table also displays the distinction between ordinary and projective domains. If $c \neq 0$, ordinary first-slot division z/c is defined on all of K. Ordinary second-slot division c/z is defined only for $z \neq 0$. Projectively, the latter sends 0 to ∞ and ∞ to 0. This is a valid action on $\mathbb{P}^1(K)$ but not an ordinary evaluation through a division by zero.

Warning 3.1 (Projective values are not ordinary values)*.* Projective continuation records a fractional linear operator and its chart transitions. It does not certify that the original arithmetic word is ordinarily admissible at a given initial value. Whenever an endpoint in K is used as an ordinary arithmetic result, ordinary admissibility must be checked separately.

3.2 Projective evaluation of a chronological history

Let $\mathrm{Hist}^{\mathrm{nd}}_K$ denote the free marked spinal history language generated by the non-degenerate elementary contexts in the table. The superscript "nd" refers to projective non-degeneracy, not to ordinary admissibility at every affine input.

Definition 3.2 (Projective evaluation)**.** Choose for each elementary non-degenerate context g_i a matrix representative $M_i \in \mathrm{GL}_2(K)$. For the chronological word $\gamma = (g_1, \ldots, g_n)$, with g_1 applied first, define

$$\rho(\gamma) = [M_n \cdots M_1] \in \mathrm{PGL}_2(K). \tag{8}$$

For the empty word, set $\rho(\varnothing) = 1$.

This definition uses projective operator evaluation. For a bounded history $(x;\gamma)$, its projective endpoint is $\rho(\gamma) \cdot x$ in $\mathbb{P}^1(K)$. Its ordinary endpoint $\nu_x(\gamma)$ remains the partial evaluation defined in Section 2.

Proposition 3.3 (Compatibility with chronology)**.** *Projective evaluation is independent of the chosen scalar matrix representatives. If $\gamma\delta$ means first γ and then δ, then*

$$\rho(\gamma\delta) = \rho(\delta)\rho(\gamma). \tag{9}$$

Moreover, whenever γ is ordinarily admissible at x and its ordinary endpoint lies in K, the projective endpoint agrees with it in the affine chart:

$$\rho(\gamma) \cdot x = [\nu_x(\gamma) : 1].$$

Proof. Replacing M_i by $a_i M_i$, with $a_i \in K^\times$, multiplies the product $M_n \cdots M_1$ by the nonzero scalar $\prod_i a_i$ and hence does not alter its projective class. If $\gamma = (g_1, \ldots, g_m)$ and $\delta = (h_1, \ldots, h_n)$, then the matrix word for $\gamma\delta$ is

$$N_n \cdots N_1 M_m \cdots M_1,$$

which represents $\rho(\delta)\rho(\gamma)$. This proves (9).

For the last assertion, each row of the table agrees with its ordinary arithmetic context wherever that context is defined. Applying this observation inductively along the admissible intermediate states proves equality in the affine chart. □

Thus ρ is a homomorphism from chronological words if the target group is written with the opposite multiplication. We retain ordinary matrix multiplication and the explicit rule (9), because it makes both the column action and function composition standard.

3.3 Generation of the projective group

Introduce the elementary projective transformations

$$T_s(z) = z + s, \qquad D_q(z) = qz \quad (q \in K^\times), \qquad J(z) = -\frac{1}{z}.$$

They are projective evaluations of, respectively, addition by s, multiplication by q, and second-slot division with numerator -1. These three kinds of context suffice for the entire bilateral projective language at the operator level.

Theorem 3.4 (Bilateral arithmetic generates PGL_2)**.** *Let K be any field. The projective evaluations of non-degenerate bilateral four-operation marked spinal histories have image*

$$\rho(\mathrm{Hist}_K^{\mathrm{nd}}) = \mathrm{PGL}_2(K).$$

Equivalently, translations T_s, nonzero scalings D_q, and J generate $\mathrm{PGL}_2(K)$.

Proof. Each of T_s, D_q, J is available in the bilateral context language, so every composition of them is in the evaluation image. Conversely, let

$$f(z) = \frac{Az + B}{Cz + D}, \qquad AD - BC \neq 0,$$

be an arbitrary element of $\mathrm{PGL}_2(K)$.

Suppose first that $C = 0$. The determinant condition then gives $A \neq 0$ and $D \neq 0$. Hence

$$f(z) = \frac{A}{D}z + \frac{B}{D} = T_{B/D} \circ D_{A/D}(z).$$

This is the projective evaluation of the chronological two-step word that first applies $D_{A/D}$ and then $T_{B/D}$.

Now suppose that $C \neq 0$. All parameters in the following expression are defined, and $(AD - BC)/C^2$ is nonzero. Direct calculation gives

$$\begin{aligned} T_{A/C} \circ D_{(AD-BC)/C^2} \circ J \circ T_{D/C}(z) &= \frac{A}{C} - \frac{AD - BC}{C^2} \frac{1}{z + D/C} \\ &= \frac{A(Cz + D) - (AD - BC)}{C(Cz + D)} \\ &= \frac{ACz + BC}{C(Cz + D)} = \frac{Az + B}{Cz + D}. \end{aligned}$$

With the fixed chronological convention, the corresponding history word is

$$\big(T_{D/C},\ J,\ D_{(AD-BC)/C^2},\ T_{A/C}\big),$$

whose entries are read from left to right in time. Thus f is again an evaluation of bilateral contexts. The two cases exhaust $\mathrm{PGL}_2(K)$.

No step uses an ordering or a characteristic assumption on K. In characteristic two the signs in J and in the determinant simplify, but the same calculation remains valid because $-1 = 1$ is still nonzero. $\square$

The proof is constructive: it gives a history of length at most four for any specified projective matrix, once arbitrary field constants are admitted as context parameters. It establishes surjectivity at the operator level, not uniqueness of a representing history. In fact, many distinct context words have the same projective evaluation.

Example 3.5 (A pole in an ordinarily inadmissible history)**.** Let $K = \mathbb{Q}$ and consider the one-step history $g[z] = 1/z$. It is a non-degenerate projective context with $\rho(g) \cdot 0 = \infty$. The bounded history $(0; g)$ has no ordinary endpoint. Appending another inversion gives a projective word acting as the identity, but the two-step ordinary evaluation at 0 is still inadmissible because its first step is undefined.

3.4 A distinguished arithmetic sublanguage: the q=4 Hecke group

The surjectivity theorem allows arbitrary constants and produces the whole projective group. A much smaller constant alphabet already carries a useful discrete geometry. Work in $\mathbb{Q}(\sqrt{2}) \subset \mathbb{R}$, and let Hist_4 be the symmetric marked history language generated by the three elementary contexts

$$\mathsf{t}_+[z] = z + \sqrt{2}, \qquad \mathsf{t}_-[z] = z - \sqrt{2}, \qquad \mathsf{j}[z] = -\frac{1}{z}.$$

Here "symmetric" means that the inverse translation is retained as a generator; j is its own projective inverse. Write

$$T = T_{\sqrt{2}}, \qquad G_4 = \rho(\mathrm{Hist}_4) = \langle T, J\rangle \le \mathrm{PGL}_2(\mathbb{R}).$$

This is the standard projective realization of the $q = 4$ Hecke group [6, 4].

Proposition 3.6 (The order-four arithmetic relation)**.** *With the column action and chronological convention fixed above, the generators have determinant-one representatives*

$$\widetilde{T} = \begin{pmatrix} 1 & \sqrt{2} \\ 0 & 1 \end{pmatrix}, \qquad \widetilde{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \textit{in } \mathrm{SL}_2(\mathbb{Z}[\sqrt{2}]).$$

Consequently G_4 is defined by matrices over $\mathbb{Z}[\sqrt{2}]$. In $\mathrm{PGL}_2(\mathbb{R})$ one has

$$J^2 = 1, \qquad (JT)^4 = 1, \tag{10}$$

and JT has exact projective order four. The second relation is the projective evaluation of the chronological eight-step history

$$\omega_4 = (\mathsf{t}_+, \mathsf{j}, \mathsf{t}_+, \mathsf{j}, \mathsf{t}_+, \mathsf{j}, \mathsf{t}_+, \mathsf{j}). \tag{11}$$

Thus ω_4 and the empty history are distinct marked histories with the same projective operator.

Proof. Both displayed matrices have determinant one and entries in $\mathbb{Z}[\sqrt{2}]$; their inverses have the same property. Moreover,

$$\widetilde{J}^2 = -I, \qquad R := \widetilde{J}\widetilde{T} = \begin{pmatrix} 0 & -1 \\ 1 & \sqrt{2} \end{pmatrix}, \qquad R^2 = \begin{pmatrix} -1 & -\sqrt{2} \\ \sqrt{2} & 1 \end{pmatrix}.$$

A second multiplication gives $R^4 = -I$. Hence the scalar classes satisfy (10). Neither R nor R^2 is scalar, so the projective order of JT is exactly four. Finally, each chronological pair $(\mathsf{t}_+, \mathsf{j})$ evaluates as $J \circ T$ and is represented by $\widetilde{J}\widetilde{T} = R$. Four such pairs therefore evaluate as R^4, which is the identity in the projective group. □

The proposition is an operator statement, not a universal statement about ordinary real arithmetic. Every letter of Hist_4 is projectively non-degenerate, but a bounded history is ordinarily admissible only if the evolving affine value is nonzero whenever j is applied. For example, ω_4 is projectively neutral but is not ordinarily admissible at $x = -\sqrt{2}$: its first translation reaches zero, after which j requests division by zero. In contrast, the same matrices act without an interior pole on the upper half-plane $\mathbb{H}^2$. The Fuchsian action and ordinary evaluation on a specified real input are therefore two related but distinct uses of the same context word.

Remark 3.7 (From histories to Hecke cells). The conventional action of G_4 on $\mathbb{H}^2$ supplies its Hecke tessellations and derived dual cellulations. Paper I needs only the following formal interface. If $\mathcal{C}$ is a chosen cell in such a G_4-equivariant cellulation and $H_{\mathcal{C}} = \mathrm{Stab}_{G_4}(\mathcal{C})$, then the cell reached by a history factors as

$$\mathrm{Hist}_4 \xrightarrow{\rho} G_4 \longrightarrow G_4/H_{\mathcal{C}}, \qquad \gamma \longmapsto \rho(\gamma)H_{\mathcal{C}}. \tag{12}$$

Neither arrow is generally injective. Relations such as (11) identify distinct histories at the operator level, and distinct operators in one coset act on $\mathcal{C}$ as the same cell. Passing instead to one endpoint uses the stabilizer of that point and forgets still more information. Thus arithmetic histories, projective operators, cells, and endpoints must not be placed in one-to-one correspondence. The order-four relation provides an arithmetic entry point for the fourfold incidence visible in the $q = 4$ geometry; the construction of analytic functions and singular zero sets on that cellulation is deferred to Papers II and III.

3.5 The affine syntactic sector

Let B_∞ be the stabilizer of the chosen point at infinity:

$$B_\infty = \mathrm{Stab}_{\mathrm{PGL}_2(K)}(\infty).$$

An element represented by $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ fixes $\infty = [1 : 0]$ exactly when $C = 0$. After rescaling by D^{-1}, it then has a unique representative

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}, \qquad a \in K^\times, \quad b \in K,$$

acting as $z \mapsto az + b$.

Define $\mathrm{Hist}_K^{\mathrm{aff}}$ to be the syntactic sublanguage generated by those *elementary non-degenerate contexts* in the table that individually fix ∞. Equivalently, it excludes the elementary second-slot divisions c/z and retains every non-degenerate elementary affine context. This definition is deliberately syntactic: it does not assert that $\mathrm{Hist}_K^{\mathrm{aff}}$ is the entire inverse image $\rho^{-1}(B_\infty)$. A word containing projective steps can leave the affine chart and later evaluate to an affine operator.

Corollary 3.8 (Affine/Borel sector)**.** *For every field K,*

$$\rho(\mathrm{Hist}_K^{\mathrm{aff}}) = B_\infty \cong \mathrm{Aff}(1, K).$$

In particular, the affine syntactic sublanguage maps surjectively onto the Borel subgroup stabilizing ∞.

Proof. Every generator of $\mathrm{Hist}_K^{\mathrm{aff}}$ fixes ∞; therefore every product of their projective matrices fixes ∞, and the image is contained in B_∞. Conversely, every element of B_∞ acts as $z \mapsto az + b$ with $a \neq 0$. It is

$$T_b \circ D_a,$$

the evaluation of a chronological word consisting of the non-degenerate affine contexts D_a followed by T_b. Hence the image is all of B_∞. The displayed normal form also identifies its group law with that of $\mathrm{Aff}(1, K)$. □

Figure 4 records the resulting operator-level placement. It does not identify the affine syntactic language with the full inverse image of the inner region.

The corollary locates the existing add–multiply language without collapsing the larger syntax. Pure slot-(1) non-degenerate histories lie in this affine sector, but so do second-slot subtraction and multiplication contexts. The criterion is stabilization of ∞, not the slot bit by itself.

3.6 The positive real component

Later differential-geometric sections use real multiplication in the form

$$z \longmapsto e^M z, \qquad M \in \mathbb{R}.$$

Its multiplier is positive. Together with all real translations, these maps form

$$\mathrm{Aff}^+(1, \mathbb{R}) = \{z \mapsto az + b : a > 0\},$$

the identity component of the real affine group. They do not include affine reflections with negative multiplier. The algebraic statement of Corollary 3.8 concerns all nonzero multipliers in $K^\times$; the continuous theory is its positive real component.

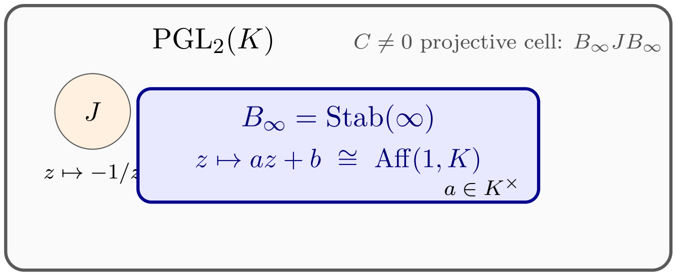

Figure 4: The affine/Borel sector inside the bilateral projective operator semantics. Elementary affine contexts generate the stabilizer B_∞, whereas inversion reaches the complementary Bruhat cell. Words containing projective steps may nevertheless evaluate back into B_∞.

3.7 Bruhat placement and the projective infinitesimal direction

The case division in the proof of Theorem 3.4 also gives the rank-one Bruhat decomposition

$$\mathrm{PGL}_2(K) = B_\infty \sqcup B_\infty J B_\infty.$$

Indeed, $C = 0$ is precisely the first cell, while the explicit decomposition for $C \neq 0$ lies in the second. This observation places bilateral arithmetic as a projective completion of the affine sector; no representation-theoretic conclusions beyond this decomposition are needed here.

There is also an infinitesimal counterpart over $\mathbb{R}$ (or $\mathbb{C}$). The projective vector fields

$$\partial_z, \qquad z\partial_z, \qquad z^2\partial_z$$

span the local projective Lie algebra. The first two generate affine flow. A conjugated translation detects the third direction:

$$J \circ T_s \circ J^{-1}(z) = \frac{z}{1 - sz} = z + sz^2 + O(s^2).$$

Consequently a general scalar projective flow has Riccati form

$$\dot{z} = \beta + \alpha z + \kappa z^2.$$

Paper I develops only the affine slice $\kappa = 0$. No projective metric, projective arithmetic expression space, or projective contact theory is asserted here.

4 Affine Cocycles and Relative Defects

We now restrict projective evaluation to the affine syntactic sector. Every evaluated history then has the form

$$x \longmapsto \Phi x + \xi, \qquad \Phi \in K^\times, \quad \xi \in K.$$

The accumulated scale Φ and translation ξ retain more of the process than an endpoint value. Their composition law forces two natural translation coordinates. The target-frame coordinate ξ weights each translation by the scales that occur after it; the source-normalized coordinate $\widehat{\xi} = \Phi^{-1}\xi$ discounts it by the scales accumulated up to that point.

Throughout Subsections 4.1– 4.4, K is an arbitrary field and every affine multiplier is nonzero. The differential calculation in Subsection 4.3 is over $\mathbb{R}$.

4.1 Discrete affine evaluation

Let

$$f_i(x) = s_i x + t_i, \qquad s_i \in K^\times, \quad t_i \in K,$$

and regard $(f_1, \dots, f_n)$ as a chronological history: f_1 acts first. Write

$$f_n \circ \cdots \circ f_1(x) = \Phi_n x + \xi_n. \tag{13}$$

For the empty history we set $\Phi_0 = 1$ and $\xi_0 = 0$.

Proposition 4.1 (Affine cocycle formulas)**.** *For a chronological affine history* $(f_1, \dots, f_n)$*,*

$$\Phi_n = \prod_{i=1}^{n} s_i, \tag{14}$$

$$\xi_n = \sum_{i=1}^{n} t_i \prod_{j=i+1}^{n} s_j. \tag{15}$$

The empty product is 1*. Defining the source-normalized translation by*

$$\widehat{\xi}_n = \Phi_n^{-1} \xi_n, \tag{16}$$

one also has

$$\widehat{\xi}_n = \sum_{i=1}^{n} \frac{t_i}{\prod_{j=1}^{i} s_j}. \tag{17}$$

Proof. We prove the first two identities simultaneously by induction on n. They hold for $n = 0$. Suppose

$$f_n \circ \cdots \circ f_1(x) = \Phi_n x + \xi_n.$$

Postcomposing chronologically by f_{n+1} gives

$$f_{n+1}(\Phi_n x + \xi_n) = s_{n+1}\Phi_n x + (s_{n+1}\xi_n + t_{n+1}).$$

Therefore

$$\Phi_{n+1} = s_{n+1}\Phi_n = \prod_{i=1}^{n+1} s_i.$$

Using the induction hypothesis for ξ_n,

$$\begin{aligned} \xi_{n+1} &= s_{n+1} \sum_{i=1}^{n} t_i \prod_{j=i+1}^{n} s_j + t_{n+1} \\ &= \sum_{i=1}^{n} t_i \prod_{j=i+1}^{n+1} s_j + t_{n+1} \prod_{j=n+2}^{n+1} s_j, \end{aligned}$$

which is exactly (15) for $n + 1$.

Every s_i is nonzero, so Φ_n is invertible. Dividing the ith term of (15) by $\Phi_n = \prod_{j=1}^{n} s_j$ gives

$$\frac{t_i \prod_{j=i+1}^{n} s_j}{\prod_{j=1}^{n} s_j} = \frac{t_i}{\prod_{j=1}^{i} s_j}.$$

Summation proves (17). ∎

Formula (15) is a future-weighting rule. A translation t_i is expressed in target coordinates only after every later scale $s_{i+1}, \dots, s_n$ has acted. Formula (17) is a past-normalization rule: the same translation is transported back through the scale accumulated from the source through step i. Neither formula is an additional geometric postulate.

Example 4.2 (Three steps)**.** For

$$f_1(x) = s_1 x + t_1, \quad f_2(x) = s_2 x + t_2, \quad f_3(x) = s_3 x + t_3,$$

direct substitution gives

$$f_3 f_2 f_1(x) = s_3 s_2 s_1 x + s_3 s_2 t_1 + s_3 t_2 + t_3.$$

Here and below juxtaposition of the displayed functions means ordinary functional composition $f_3 \circ f_2 \circ f_1$, not chronological word concatenation. The formula makes the future scales attached to each translation visible.

4.2 Cocycle laws under chronological concatenation

For an affine history γ, write

$$\rho(\gamma)(x) = \Phi_\gamma x + \xi_\gamma, \qquad \widehat{\xi}_\gamma = \Phi_\gamma^{-1} \xi_\gamma.$$

Recall that $\gamma\delta$ means first γ and then δ, so $\rho(\gamma\delta) = \rho(\delta)\rho(\gamma)$.

Proposition 4.3 (Chronological affine cocycle law)**.** *For affine histories γ and δ,*

$$\Phi_{\gamma\delta} = \Phi_\delta \Phi_\gamma, \tag{18}$$

$$\xi_{\gamma\delta} = \Phi_\delta \xi_\gamma + \xi_\delta, \tag{19}$$

$$\widehat{\xi}_{\gamma\delta} = \widehat{\xi}_\gamma + \Phi_\gamma^{-1} \widehat{\xi}_\delta. \tag{20}$$

If γ is invertible, then

$$\Phi_{\gamma^{-1}} = \Phi_\gamma^{-1}, \qquad \xi_{\gamma^{-1}} = -\Phi_\gamma^{-1} \xi_\gamma = -\widehat{\xi}_\gamma. \tag{21}$$

Proof. Apply $\rho(\delta)$ to $\rho(\gamma)(x)$:

$$\rho(\delta)(\Phi_\gamma x + \xi_\gamma) = \Phi_\delta \Phi_\gamma x + \Phi_\delta \xi_\gamma + \xi_\delta.$$

This proves (18) and (19). Since K is commutative,

$$\begin{aligned} \widehat{\xi}_{\gamma\delta} &= (\Phi_\delta \Phi_\gamma)^{-1} (\Phi_\delta \xi_\gamma + \xi_\delta) \\ &= \Phi_\gamma^{-1} \xi_\gamma + \Phi_\gamma^{-1} \Phi_\delta^{-1} \xi_\delta, \end{aligned}$$

which is (20). Finally, solving $y = \Phi_\gamma x + \xi_\gamma$ for x yields

$$\rho(\gamma)^{-1}(y) = \Phi_\gamma^{-1} y - \Phi_\gamma^{-1} \xi_\gamma,$$

and hence (21). □

These formulas show why the translation coordinate is a cocycle rather than an additive homomorphism. The semidirect-product action of scale on translation is part of the arithmetic composition law.

4.3 Left and right affine differentials

We now pass to the positive real affine group and write

$$g(M,\xi) = \begin{pmatrix} e^M & \xi \\ 0 & 1 \end{pmatrix}. \tag{22}$$

The symbol M denotes accumulated logarithmic scale; it is distinct from the fixed infinitesimal intensity λ used in later flow equations.

Proposition 4.4 (Left and right affine differentials)**.** *For the matrix g in (22), the translation components of the left and right Maurer–Cartan forms are*

$$\left(g^{-1}dg\right)_{\mathrm{tr}} = e^{-M}\, d\xi, \tag{23}$$

$$\left(dg\, g^{-1}\right)_{\mathrm{tr}} = d\xi - \xi\, dM. \tag{24}$$

If $\widehat{\xi} = e^{-M}\xi$, then

$$d\xi - \xi\, dM = e^M d\widehat{\xi}. \tag{25}$$

Proof. We have

$$g^{-1} = \begin{pmatrix} e^{-M} & -e^{-M}\xi \\ 0 & 1 \end{pmatrix}, \qquad dg = \begin{pmatrix} e^M dM & d\xi \\ 0 & 0 \end{pmatrix}.$$

Multiplication on the two sides gives

$$g^{-1}dg = \begin{pmatrix} dM & e^{-M}d\xi \\ 0 & 0 \end{pmatrix}$$

and

$$dg\, g^{-1} = \begin{pmatrix} dM & d\xi - \xi dM \\ 0 & 0 \end{pmatrix}.$$

The upper-right entries prove (23) and (24). Differentiating $\widehat{\xi} = e^{-M}\xi$ gives

$$d\widehat{\xi} = e^{-M}(d\xi - \xi dM),$$

which is equivalent to (25). □

The left Maurer–Cartan form $g^{-1}dg$ measures velocity in body or source coordinates; the right form $dg\, g^{-1}$ measures it in spatial or target coordinates. This left–right group distinction is not the same as operand-slot chirality. Relating a slot choice to left or right group multiplication requires an additional rule specifying how a new context is inserted into a represented history. No such identification is assumed here.

There are likewise two separate coordinate statements. The matrix coordinate ξ is the target-frame translation in $x \mapsto e^M x + \xi$, whereas $\widehat{\xi} = e^{-M}\xi$ is its source-normalized value. Equations (23)– (25) relate their differentials but do not make the two coordinates interchangeable.

4.4 Relative affine defect

The affine cocycle permits a comparison of two histories before any global torsion theory is introduced.

Definition 4.5 (Relative affine defect)**.** For two affine histories γ and δ, the *relative affine defect of* γ *with reference* δ is

$$\mathcal{K}_{\mathrm{aff}}(\gamma,\delta) = \rho(\delta)^{-1}\rho(\gamma) \in \mathrm{Aff}(1,K). \tag{26}$$

The pair is *scale-compatible* when $\Phi_\gamma = \Phi_\delta$.

The order of the arguments in (26) is part of the convention. Interchanging γ and δ inverts the group-valued defect.

Proposition 4.6 (Coordinates of the relative affine defect)**.** *For arbitrary affine histories* γ *and* δ*,*

$$\mathcal{K}_{\mathrm{aff}}(\gamma,\delta)(x) = \frac{\Phi_\gamma}{\Phi_\delta}x + \frac{\xi_\gamma - \xi_\delta}{\Phi_\delta}. \tag{27}$$

If the pair is scale-compatible with common scale Φ*, then*

$$\nu_x(\gamma) - \nu_x(\delta) = \xi_\gamma - \xi_\delta, \tag{28}$$

$$\mathcal{K}_{\mathrm{aff}}(\gamma,\delta)(x) = x + \frac{\xi_\gamma - \xi_\delta}{\Phi}. \tag{29}$$

In particular, the endpoint difference is then independent of x*.*

Proof. The inverse of $y \mapsto \Phi_\delta y + \xi_\delta$ is $y \mapsto \Phi_\delta^{-1}(y - \xi_\delta)$. Applying it after $x \mapsto \Phi_\gamma x + \xi_\gamma$ gives (27). If $\Phi_\gamma = \Phi_\delta = \Phi$, direct subtraction of the two endpoint formulas gives (28), and substituting the common scale into (27) gives (29). □

For unequal scales, the endpoint difference contains $(\Phi_\gamma - \Phi_\delta)x$ and depends on the initial value. In that case the group-valued comparison remains defined, but there is no base-point-independent scalar translation defect of the form (28).

Example 4.7 (The elementary affine order defect)**.** Fix $\mu \in K$ and $q \in K^\times$. Let γ first add μ and then scale by q, while δ first scales by q and then adds μ. Then

$$\rho(\gamma)(x) = q(x+\mu) = qx + q\mu, \qquad \rho(\delta)(x) = qx + \mu.$$

The histories have the same scale q, and hence

$$\nu_x(\gamma) - \nu_x(\delta) = \mu(q-1), \qquad \mathcal{K}_{\mathrm{aff}}(\gamma,\delta)(x) = x + \mu(1 - q^{-1}). \tag{30}$$

Over the positive real group, writing $q = e^M$ gives the endpoint defect $\mu(e^M - 1)$.

Equation (30) is an open comparison of two chronological histories. It is not a closed commutator-loop holonomy, a comparison with path inverse, or a mirror invariant. Section 8 will place compatible two-history defects in the global torsion framework.

4.5 Propagation of a change in the initial accumulator

The accumulated scale also controls sensitivity to the initial value.

Proposition 4.8 (Exact affine perturbation propagation)**.** *Let γ be an affine history with scale Φ_γ. For any $x, \widetilde{x} \in K$,*

$$\nu_{\widetilde{x}}(\gamma) - \nu_x(\gamma) = \Phi_\gamma(\widetilde{x} - x). \tag{31}$$

When $\widetilde{x} \neq x$,

$$\frac{\nu_{\widetilde{x}}(\gamma) - \nu_x(\gamma)}{\widetilde{x} - x} = \Phi_\gamma. \tag{32}$$

Proof. The affine normal form gives

$$\nu_{\widetilde{x}}(\gamma) = \Phi_\gamma \widetilde{x} + \xi_\gamma, \qquad \nu_x(\gamma) = \Phi_\gamma x + \xi_\gamma.$$

Subtracting cancels the translation cocycle and proves (31). Division by the nonzero element $\widetilde{x} - x$ proves (32). □

This is an exact discrete identity, not a linear approximation. The continuous affine flow of the next section is obtained by differentiating the same semidirect-product action. Supplementary expansions and convention checks appear in Appendix B.

5 Continuous Arithmetic Flow

The affine cocycles of the preceding section have an infinitesimal counterpart. Translation and dilation generate the Lie algebra of the positive affine group, and their linear combinations give the continuous propagation law. The metric formulation introduced below is intrinsic: the symbols u and v label arithmetic directions, but need not be coordinates.

5.1 The affine Lie algebra and its flow

On the assignment line with coordinate z, let

$$E = \partial_z, \qquad H = z\partial_z.$$

These are respectively the infinitesimal generators of translation and positive dilation, and they satisfy $[E, H] = E$. Fix constants $\mu, \lambda \in \mathbb{R}$, with $\mu \neq 0$. The generator selected by an angle θ is

$$\Omega_\theta = \mu \cos\theta\, E + \lambda \sin\theta\, H. \tag{33}$$

Consequently, an integral curve $z = z(s)$ satisfies

$$\frac{dz}{ds} = \mu\cos\theta + \lambda z \sin\theta. \tag{34}$$

This Lie-algebra calculation is the primary derivation of the continuous AEG equation. It also fixes the convention: the translation coefficient is $\mu\cos\theta$, while the dilation coefficient is $\lambda \sin\theta$.

When θ is constant, set $b = \lambda\sin\theta$. The solution through $z(0) = z_0$ is

$$z(s) = \begin{cases} e^{bs} z_0 + \mu\cos\theta\, \dfrac{e^{bs} - 1}{b}, & b \neq 0, \\[2ex] z_0 + \mu s \cos\theta, & b = 0. \end{cases} \tag{35}$$

In particular, the positive arithmetic axes give

$$\theta = 0: \quad z(s) = z_0 + \mu s, \qquad \theta = \frac{\pi}{2}: \quad z(s) = e^{\lambda s} z_0.$$

The opposite axes give subtraction and inverse dilation.

5.2 The Pfaffian propagation constraint

There is a useful three-dimensional local state space with coordinates (u, v, z). On it define

$$\alpha = dz - \mu\, du - \lambda z\, dv \tag{36}$$

and the two horizontal vector fields

$$D_u = \partial_u + \mu\partial_z, \qquad D_v = \partial_v + \lambda z\partial_z. \tag{37}$$

A curve tangent to $\cos\theta\, D_u + \sin\theta\, D_v$ satisfies (34). Thus the frequently used notation

$$dz = \mu\, du + \lambda z\, dv$$

means that α vanishes on admissible tangent vectors. It is not an identity asserting that z is a function of two independent coordinates whose ordinary partial derivatives are simultaneously μ and λz. Indeed,

$$[D_u, D_v] = \mu\lambda\, \partial_z. \tag{38}$$

When $\mu\lambda \neq 0$, this bracket already rules out such an integrable coordinate interpretation.

5.3 Regular arithmetic expression spaces

Definition 5.1 (Regular arithmetic expression space)**.** A *regular arithmetic expression space*, or regular AES, is a tuple

$$\mathfrak{E} = (M, g, a; \mu, \lambda)$$

such that M is an oriented smooth surface, possibly with boundary, g is a smooth Riemannian metric, $a \in C^\infty(M, \mathbb{R})$, $\mu \in \mathbb{R} \setminus \{0\}$, $\lambda \in \mathbb{R}$, and

$$|\nabla a|_g^2 = \mu^2 + \lambda^2 a^2 \tag{39}$$

everywhere on M. The function a is called the *assignment function.*

The metric in Definition 5.1 is part of the structure. Equation (39) is therefore an eikonal equation on a specified Riemannian surface, not a metric-free identity.

Let J denote rotation by $+\pi/2$ in the oriented tangent planes. Since $\mu \neq 0$, the quantity

$$q = \sqrt{\mu^2 + \lambda^2 a^2}$$

is everywhere positive. The next proposition shows that the eikonal definition contains exactly the oriented orthonormal directional data used in the flow equation.

Proposition 5.2 (Canonical arithmetic frame)**.** *For an oriented Riemannian surface (M, g), a smooth function a, and constants $\mu \neq 0$ and λ, the following are equivalent.*

1. *The eikonal identity (39) holds.*

2. *There is a unique positively oriented orthonormal frame (X_u, X_v) satisfying*

$$X_u a = \mu, \qquad X_v a = \lambda a. \tag{40}$$

When these conditions hold, put $E_a = \nabla a/q$. The frame is

$$\boxed{\begin{aligned} X_u &= \frac{\mu}{q}E_a - \frac{\lambda a}{q}JE_a, \\ X_v &= \frac{\lambda a}{q}E_a + \frac{\mu}{q}JE_a. \end{aligned}} \tag{41}$$

Proof. Assume the eikonal identity. Then $|\nabla a| = q > 0$, so E_a and JE_a form a smooth positively oriented orthonormal frame. The coefficient matrix in (41) is

$$\frac{1}{q}\begin{pmatrix} \mu & \lambda a \\ -\lambda a & \mu \end{pmatrix},$$

whose columns are orthonormal and whose determinant is one. Hence (X_u, X_v) is positively oriented and orthonormal. Since $da(E_a) = q$ and $da(JE_a) = 0$, the two identities in (40) follow.

For uniqueness, write any positively oriented orthonormal frame in the form

$$X_u = cE_a + sJE_a, \qquad X_v = -sE_a + cJE_a, \qquad c^2 + s^2 = 1.$$

The two directional identities force $qc = \mu$ and $-qs = \lambda a$. Thus $c = \mu/q$ and $s = -\lambda a/q$, which gives (41).

Conversely, if an orthonormal frame satisfies (40), then

$$|\nabla a|^2 = (X_u a)^2 + (X_v a)^2 = \mu^2 + \lambda^2 a^2.$$

□

The frame is generally non-coordinate. Applying its bracket to the assignment function gives

$$[X_u, X_v]a = X_u(\lambda a) - X_v(\mu) = \mu\lambda. \tag{42}$$

Thus, when $\lambda \neq 0$, the fields cannot commute. If (η^u, η^v) is their dual coframe, the valid intrinsic Pfaffian statement is

$$da = \mu\, \eta^u + \lambda a\, \eta^v;$$

the one-forms η^u, η^v are not presumed exact.

Theorem 5.3 (Continuous affine flow)**.** *Let $\mathfrak{E} = (M, g, a; \mu, \lambda)$ be a regular AES with its canonical arithmetic frame. If a unit-speed curve γ has tangent*

$$\dot\gamma = \cos\theta\, X_u + \sin\theta\, X_v,$$

where θ may vary along the curve, then

$$\frac{d}{ds}(a \circ \gamma) = \mu\cos\theta + \lambda(a \circ \gamma)\sin\theta. \tag{43}$$

For constant θ, its solutions are given by (35).

Proof. By the chain rule and (40),

$$\frac{d}{ds}(a \circ \gamma) = da(\dot\gamma) = \cos\theta\, X_u a + \sin\theta\, X_v a = \mu\cos\theta + \lambda a \sin\theta.$$

For constant θ, this is the linear equation already integrated in (35). □

5.4 Rectification and the local order defect

For $\lambda \neq 0$, define

$$r = \operatorname{arcsinh}\left(\frac{\lambda a}{\mu}\right). \tag{44}$$

The chain rule and (39) give

$$|\nabla r| = |\lambda|. \tag{45}$$

Thus r has constant gradient magnitude. When $\lambda = 0$, the original equation already reduces to $|\nabla a| = |\mu|$, and a/μ is a unit-gradient rectifying function.

The same affine convention fixes the sign of the elementary local order defect. For small increments h and k, let

$$A_h(z) = z + \mu h, \qquad M_k(z) = e^{\lambda k} z.$$

Then

$$\begin{aligned}(M_k \circ A_h)(z) - (A_h \circ M_k)(z) &= \mu h(e^{\lambda k} - 1) \\ &= \mu\lambda\, hk + O\Big(\|(h,k)\|^3\Big).\end{aligned} \tag{46}$$

This is an infinitesimal two-order comparison, not an exact area formula for an arbitrary finite region.

5.5 Affine flow inside projective flow

An infinitesimal projective transformation of the line has the Riccati form

$$\dot{z} = \beta + \alpha z + \kappa z^2.$$

The theory developed in this section is the affine slice $\kappa = 0$, with $\beta = \mu \cos\theta$ and $\alpha = \lambda \sin\theta$. Results proved for regular affine AESs below do not automatically extend to the quadratic projective flow.

6 The Basic Hyperbolic Expression Space

The regular-AES axioms admit a homogeneous realization on the positive affine group. This construction explains the upper-half-plane metric from the group action and the normalization of the arithmetic frame; it does not assert that every regular AES is homogeneous or that the model is unique.

6.1 The positive affine group

Write

$$G = \mathrm{Aff}^+(1,\mathbb{R}) = \{(x,y) : x \in \mathbb{R},\ y > 0\}$$

with multiplication

$$(x,y)(x',y') = (x + yx', yy'). \tag{47}$$

The element (x,y) represents the affine map $t \mapsto x + yt$. Left translation by (x_0, y_0) is

$$L_{(x_0,y_0)}(x,y) = (x_0 + y_0 x, y_0 y).$$

For fixed $\mu, \lambda > 0$, consider the left-invariant vector fields

$$X_u = -\mu y\,\partial_x, \qquad X_v = -\lambda y\,\partial_y. \tag{48}$$

Both signs are chosen so that the frame is positively oriented for $dx \wedge dy$ and has the desired action on the assignment below.

Proposition 6.1 (Invariant affine metric)**.** *The Riemannian metric for which the frame* (48) *is orthonormal is*

$$g_{\mu,\lambda} = \frac{1}{y^2}\left(\frac{dx^2}{\mu^2} + \frac{dy^2}{\lambda^2}\right). \tag{49}$$

It is invariant under every left translation of G.

Proof. The coframe dual to (X_u, X_v) is

$$-\frac{dx}{\mu y}, \qquad -\frac{dy}{\lambda y},$$

whose sum of squares is (49). Under $L_{(x_0,y_0)}$, one has

$$x \longmapsto x_0 + y_0 x, \qquad y \longmapsto y_0 y, \qquad dx \longmapsto y_0 dx, \qquad dy \longmapsto y_0 dy.$$

Substitution in (49) shows $L^*_{(x_0,y_0)} g_{\mu,\lambda} = g_{\mu,\lambda}$. □

6.2 The basic hyperbolic arithmetic expression space

Let

$$\mathbb{H}^2 = \{(x,y) \in \mathbb{R}^2 : y > 0\}$$

with its standard orientation, and define

$$a(x,y) = -\frac{x}{y}. \tag{50}$$

Theorem 6.2 (Basic hyperbolic AES)**.** *For every $\mu, \lambda > 0$, the tuple*

$$\mathfrak{E}_{\mathrm{hyp}}(\mu,\lambda) = (\mathbb{H}^2, g_{\mu,\lambda}, a; \mu, \lambda)$$

is a regular arithmetic expression space. Its canonical arithmetic frame is the left-invariant frame (48)*. Moreover, $(\mathbb{H}^2, g_{\mu,\lambda})$ is complete and has constant Gaussian curvature*

$$K = -\lambda^2.$$

Proof. Direct differentiation gives

$$a_x = -\frac{1}{y}, \qquad a_y = \frac{x}{y^2} = -\frac{a}{y}.$$

Therefore the frame (48) satisfies

$$X_u a = \mu, \qquad X_v a = \lambda a.$$

It is positively oriented and orthonormal by Proposition 6.1. The frame equivalence of Proposition 5.2 now proves the regular-AES identity. Equivalently, using

$$g^{-1}_{\mu,\lambda} = \mu^2 y^2\,\partial_x \otimes \partial_x + \lambda^2 y^2\,\partial_y \otimes \partial_y,$$

one obtains directly

$$|\nabla a|^2_{g_{\mu,\lambda}} = \mu^2 y^2 a_x^2 + \lambda^2 y^2 a_y^2 = \mu^2 + \lambda^2 a^2.$$

For completeness and curvature, introduce the global coordinate

$$X = \frac{\lambda}{\mu} x.$$

Then

$$g_{\mu,\lambda} = \frac{1}{\lambda^2} \frac{dX^2 + dy^2}{y^2}. \tag{51}$$

The metric in parentheses is the complete standard upper-half-plane metric of curvature -1. Multiplication of a Riemannian metric by λ^{-2} preserves completeness and multiplies Gaussian curvature by λ^2. Hence $g_{\mu,\lambda}$ is complete and has curvature $-\lambda^2$. □

The parameters have different geometric roles. The change $X = (\lambda/\mu)x$ removes μ from the metric, so μ fixes the additive normalization of the assignment frame. The parameter λ sets the curvature scale. No uniqueness claim for regular AESs follows from this normalization.

6.3 Horocyclic coordinates and assignment contours

Set $y = e^{\lambda v}$ and $u = x$. Then

$$g_{\mu,\lambda} = e^{-2\lambda v} \frac{du^2}{\mu^2} + dv^2, \qquad a(u,v) = -ue^{-\lambda v}. \tag{52}$$

The curves $y =$ constant are horocycles based at the ideal point ∞, and the vertical lines $x =$ constant are their orthogonal geodesics. The assignment contour of value c is

$$a^{-1}(c) = \{x = -cy,\ y > 0\}.$$

In particular, the zero locus in this model is the vertical geodesic $\{x = 0,\ y > 0\}$.

6.4 The Laplace eigenfunction

We use the divergence-of-gradient sign convention

$$\Delta_g f = \frac{1}{\sqrt{\det g}} \partial_i \Big(\sqrt{\det g}\, g^{ij} \partial_j f \Big). \tag{53}$$

Proposition 6.3 (Laplace eigenfunction)**.** *The assignment function of the basic hyperbolic AES satisfies*

$$\Delta_{g_{\mu,\lambda}} a = 2\lambda^2 a. \tag{54}$$

Proof. Since

$$\sqrt{\det g_{\mu,\lambda}} = \frac{1}{\mu\lambda y^2},$$

the products $\sqrt{\det g}\, g^{xx} = \mu/\lambda$ and $\sqrt{\det g}\, g^{yy} = \lambda/\mu$ are constant. Thus

$$\Delta_{g_{\mu,\lambda}} f = \mu^2 y^2 f_{xx} + \lambda^2 y^2 f_{yy}.$$

For $a = -x/y$, one has $a_{xx} = 0$ and $a_{yy} = -2x/y^3 = 2a/y^2$. Substitution gives (54). □

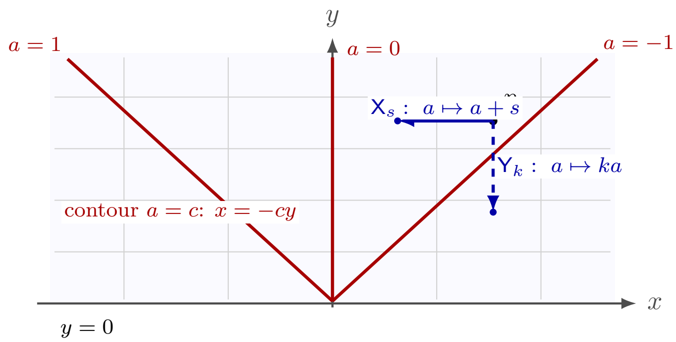

Figure 5: The basic hyperbolic arithmetic grid on the upper half-plane. Gray lines show the horocycle–geodesic background; red rays are the level sets $a = -x/y$. The blue motions implement addition and positive multiplication on the assignment. They preserve the indicated assignment rules but are not, in general, isometries of $g_{\mu,\lambda}$.

6.5 Assignment-compatible grid transformations

For $s \in \mathbb{R}$ and $k > 0$, define

$$\mathsf{X}_s(x, y) = (x - sy, y), \qquad \mathsf{Y}_k(x, y) = \left(x, \frac{y}{k}\right). \tag{55}$$

They implement addition and positive multiplication at the level of the assignment:

$$a \circ \mathsf{X}_s = a + s, \qquad a \circ \mathsf{Y}_k = ka. \tag{56}$$

Figure 5 separates the assignment contours from the horocyclic background and also marks the two assignment-compatible motions. The pullback calculation following the figure determines their metric status.

These maps are *assignment-compatible transformations*; in general they are not isometries of $g_{\mu,\lambda}$. Indeed,

$$\mathsf{X}_s^* g_{\mu,\lambda} = \frac{1}{y^2}\left(\frac{(dx - s\,dy)^2}{\mu^2} + \frac{dy^2}{\lambda^2}\right), \tag{57}$$

$$\mathsf{Y}_k^* g_{\mu,\lambda} = \frac{1}{y^2}\left(\frac{k^2 dx^2}{\mu^2} + \frac{dy^2}{\lambda^2}\right). \tag{58}$$

Consequently X_s is an isometry only for $s = 0$, and Y_k only for $k = 1$.

Remark 6.4 (A Baumslag–Solitar relation). For an integer $n \geq 2$, $\mathsf{X}_s^n = \mathsf{X}_{ns}$, and direct composition gives

$$\mathsf{Y}_n^{-1} \circ \mathsf{X}_s^n \circ \mathsf{Y}_n = \mathsf{X}_s. \tag{59}$$

With stable letter $T = \mathsf{Y}_n^{-1}$, this reads $T\mathsf{X}_s^n T^{-1} = \mathsf{X}_s$, a Baumslag–Solitar relation in the $BS(n, 1)$ convention. Equation (59) is an identity among assignment-compatible diffeomorphisms, not a claim that the arithmetic grid acts by hyperbolic isometries. No complexity conclusion is inferred from it.

Coordinate derivations of the group chart, gradient, curvature, Laplacian, metric pullbacks, and relation (59) are collected in Appendix C.

7 Zero Loci and Singular Expression Spaces

The eikonal equation imposes an immediate restriction on the zero set of a regular assignment: its gradient cannot vanish there. This separates regular zero geometry from isolated zeros, crossings, and the parameter-dependent singular phenomena reserved for later work.

7.1 Regular zero loci

For an assignment function a, write

$$Z(a) = a^{-1}(0).$$

Theorem 7.1 (Regular zero locus)**.** *Let $\mathfrak{E} = (M, g, a; \mu, \lambda)$ be a regular AES.*

1. *If M has no boundary, then $Z(a)$ is a smooth embedded codimension-one submanifold of M.*
2. *If M has boundary, then $Z(a) \cap \operatorname{int} M$ is a smooth embedded codimension-one submanifold of $\operatorname{int} M$.*

In particular, for a surface the indicated zero locus is a disjoint union of smooth one-dimensional components. No assertion is made here about finiteness, properness, compactness, or behavior at ∂M.

Proof. At every $p \in Z(a)$, the defining eikonal identity gives

$$|\nabla a(p)|_g^2 = \mu^2.$$

Because $\mu \neq 0$, one has $da_p \neq 0$. Thus 0 is a regular value of a when M is boundaryless, and of $a|_{\operatorname{int} M}$ for interior zeros when a boundary is present. The regular-value theorem gives the stated embedded submanifolds. □

Corollary 7.2 (Local rigidity of regular zeros)**.** *Within the interior of a regular two-dimensional AES, the zero locus has no isolated point, crossing, branch point, or endpoint. None of these configurations can occur as the germ of $Z(a)$ at an interior point.*

Proof. By Theorem 7.1, a neighborhood of every interior zero admits smooth coordinates in which $a^{-1}(0)$ is a coordinate line. Such a germ has none of the listed configurations. □

This conclusion is local. Without additional global hypotheses, it does not classify the number or topology of the components. In the basic hyperbolic model, for example, $Z(a) = \{x = 0, y > 0\}$. Completeness and the absence of a boundary strengthen the conclusion substantially.

7.2 Complete regular splitting and a global no-go theorem

Theorem 7.3 (Complete regular AES splitting)**.** *Let $\mathfrak{E} = (M, g, a; \mu, \lambda)$ be a regular AES for which M is connected, has no boundary, and is complete as a Riemannian manifold. Define*

$$\mathcal{R}(t) = \int_0^t \frac{ds}{\sqrt{\mu^2 + \lambda^2 s^2}}, \qquad r = \mathcal{R} \circ a. \tag{60}$$

Then $|\nabla r|_g = 1$, the vector field $X = \nabla r$ has a global flow $\Phi : \mathbb{R} \times M \to M$, and

$$F : Z(a) \times \mathbb{R} \longrightarrow M, \qquad F(q, t) = \Phi_t(q), \tag{61}$$

is a diffeomorphism. In particular, a is surjective, $Z(a)$ is nonempty and connected, and M is noncompact.

The conclusion is a smooth product decomposition. It does not assert that F is an isometry from a Riemannian product.

Proof. Put $q(a) = \sqrt{\mu^2 + \lambda^2 a^2}$. Since $\mu \neq 0$, this is everywhere positive. The chain rule and the regular-AES equation give

$$\nabla r = \frac{\nabla a}{q(a)}, \qquad |\nabla r|_g^2 = \frac{|\nabla a|_g^2}{q(a)^2} = 1.$$

Thus $X = \nabla r$ is a smooth bounded vector field. To see that its flow is complete, let an integral curve be defined on a finite time interval. Its speed is one, so over every bounded time interval it remains in a closed metric ball. By Hopf–Rinow, closed bounded balls in the complete connected Riemannian manifold M are compact. The standard local existence theorem for smooth vector fields then extends the integral curve past every finite endpoint. Hence Φ_t exists for all $t \in \mathbb{R}$.

Along this flow,

$$\frac{d}{dt} r(\Phi_t(p)) = dr(X) = |\nabla r|_g^2 = 1, \qquad r(\Phi_t(p)) = r(p) + t. \tag{62}$$

It follows that every orbit meets $r^{-1}(0) = a^{-1}(0)$ exactly once, namely at $\Phi_{-r(p)}(p)$. Therefore the inverse of (61) is

$$p \longmapsto \big(\Phi_{-r(p)}(p), r(p)\big).$$

Both maps are smooth, so F is a diffeomorphism. Equation (62) also shows that $r(M) = \mathbb{R}$. The function $\mathcal{R} : \mathbb{R} \to \mathbb{R}$ is a strictly increasing diffeomorphism (linear when $\lambda = 0$, and an inverse-hyperbolic-sine rescaling when $\lambda \neq 0$); hence a is surjective. Finally, $Z(a) \times \mathbb{R} \cong M$ is connected only if $Z(a)$ is connected, and the displayed product cannot be compact. □

Remark 7.4 (Why the global hypotheses matter). If a boundary is present, the unit vector field ∇r may reach it in finite time. If the metric is incomplete, an integral curve may escape the manifold in finite time. In either case the global flow argument fails, so the theorem must not be applied to domains with boundary or to incomplete regular models without a separate extension analysis.

Theorem 7.3 is a global obstruction, not merely a local regular-value statement. A connected complete boundaryless regular scalar AES cannot have several zero components, let alone a branching or four-valent zero network. Consequently, a Hecke cellulation used as a zero graph must enter through singular points, a failure of completeness or boundarylessness, or data richer than one regular scalar assignment. Paper I records this no-go boundary; the corresponding singular constructions belong to Paper III.

Figure 6 summarizes the categorical distinction. It illustrates the local conclusions already proved; it does not classify singular zero germs.

7.3 Singular arithmetic expression spaces

The preceding theorem makes a singular framework unavoidable if one wishes to admit isolated or branching zeros.

Definition 7.5 (Singular arithmetic expression space)**.** A *singular arithmetic expression space* is a tuple

$$\mathfrak{E}_{\text{sing}} = (M, S, g, a; \mu, \lambda)$$

with the following data:

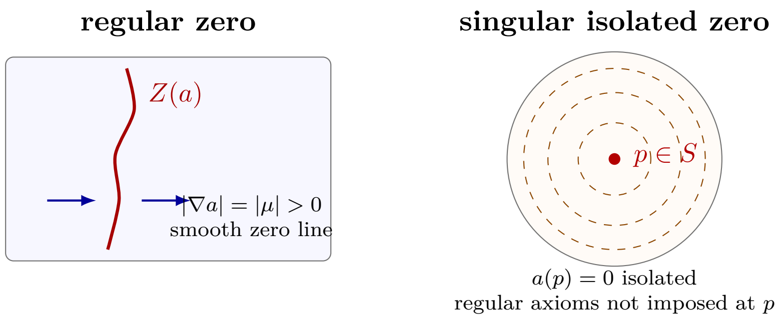

Figure 6: Regular and singular zero geometry. In a regular AES with $\mu \neq 0$, the assignment crosses zero with nonvanishing gradient, so the interior zero germ is a smooth curve. The radial disc model instead has an isolated zero at its declared singular point, where the assignment is not C^1 and the regular structure has no local extension.

1. M is a nonempty oriented smooth surface, possibly with boundary, and $S \subset M$ is a closed nowhere-dense singular set;
2. g is a smooth Riemannian metric on $M \setminus S$;
3. $a : M \to \mathbb{R}$ is a specified function whose restriction to $M \setminus S$ is smooth;
4. $\mu \in \mathbb{R} \setminus \{0\}$, $\lambda \in \mathbb{R}$, and
$$(M \setminus S, g, a|_{M \setminus S}; \mu, \lambda)$$
is a regular AES.
5. Every $p \in S$ is locally essential: there is no neighborhood U of p on which $g|_{U \setminus S}$ and $a|_{U \setminus S}$ admit smooth extensions $\widetilde{g}$ and $\widetilde{a}$, with $\widetilde{g}$ nondegenerate, such that $(U, \widetilde{g}, \widetilde{a}; \mu, \lambda)$ is a regular AES.

The metric or assignment may possess weaker extensions across S; their precise regularity and local behavior must be checked in each example.

The set S need not consist entirely of zeros. Nowhere density keeps the regular locus nonempty and dense, while local essentiality prevents an arbitrary enlargement of S by points at which the regular structure extends. For a singular AES define

$$Z_{\mathrm{reg}}(a) = Z(a) \cap (M \setminus S), \qquad Z_{\mathrm{sing}}(a) = Z(a) \cap S. \tag{63}$$

Every point of $Z_{\mathrm{reg}}(a)$ is a regular zero by Theorem 7.1; a point of $Z_{\mathrm{sing}}(a)$ is called singular because the regular structure is not asserted there, even when some of the underlying data happen to extend smoothly.

7.4 A verified isolated-zero singular model

Let

$$\mathbb{D} = \{(\xi, \eta) \in \mathbb{R}^2 : \xi^2 + \eta^2 < 1\}, \qquad \rho = \sqrt{\xi^2 + \eta^2},$$

and fix $\mu, \lambda > 0$. Equip the disc with

$$g_{\mathbb{D}} = \frac{4}{\lambda^2 (1 - \rho^2)^2} (d\xi^2 + d\eta^2) \tag{64}$$

and define

$$a_{\mathbb{D}}(\xi,\eta) = \begin{cases} \dfrac{2\mu}{\lambda}\dfrac{\rho}{1-\rho^2}, & (\xi,\eta)\neq(0,0), \\ 0, & (\xi,\eta)=(0,0). \end{cases} \tag{65}$$

Proposition 7.6 (Classification of the isolated-zero disc)**.** *The tuple*

$$(\mathbb{D}, \{0\}, g_{\mathbb{D}}, a_{\mathbb{D}}; \mu, \lambda)$$

is a singular AES with the following properties.

1. *The metric $g_{\mathbb{D}}$ extends smoothly and nondegenerately across the center.*
2. *The assignment $a_{\mathbb{D}}$ is smooth on $\mathbb{D}\setminus\{0\}$ but is not C^1 at the center.*
3. *The punctured disc is a regular AES:*

$$|\nabla a_{\mathbb{D}}|^2_{g_{\mathbb{D}}} = \mu^2 + \lambda^2 a_{\mathbb{D}}^2.$$

4. *Its zero decomposition is*

$$Z_{\mathrm{reg}}(a_{\mathbb{D}}) = \varnothing, \qquad Z_{\mathrm{sing}}(a_{\mathbb{D}}) = \{0\}.$$

Proof. Formula (64) is written in Cartesian coordinates. Its conformal factor is a positive smooth function on all of $\mathbb{D}$, proving the first claim.

Away from the origin, ρ and hence $a_{\mathbb{D}}$ are smooth. On the ξ-axis, however,

$$\frac{a_{\mathbb{D}}(t,0)-a_{\mathbb{D}}(0,0)}{t} = \frac{2\mu}{\lambda}\frac{|t|}{t(1-t^2)}.$$

The one-sided limits are $2\mu/\lambda$ and $-2\mu/\lambda$. Thus even the partial derivative in the ξ-direction does not exist at the center, so the assignment is not C^1 there. Moreover, the punctured assignment tends to zero at the center, so its continuous extension is unique. The same calculation therefore rules out every smooth local extension agreeing with the punctured assignment and proves that the declared singular point is locally essential.

On the punctured disc the assignment is radial, and

$$\frac{da_{\mathbb{D}}}{d\rho} = \frac{2\mu}{\lambda}\frac{1+\rho^2}{(1-\rho^2)^2}.$$

The inverse radial metric coefficient is

$$g_{\mathbb{D}}^{\rho\rho} = \frac{\lambda^2(1-\rho^2)^2}{4}.$$

Consequently,

$$\begin{aligned} |\nabla a_{\mathbb{D}}|^2_{g_{\mathbb{D}}} &= g_{\mathbb{D}}^{\rho\rho}\left(\frac{da_{\mathbb{D}}}{d\rho}\right)^2 \\ &= \mu^2\frac{(1+\rho^2)^2}{(1-\rho^2)^2} \\ &= \mu^2 + \lambda^2\left(\frac{2\mu}{\lambda}\frac{\rho}{1-\rho^2}\right)^2, \end{aligned}$$

which is the regular-AES equation on $\mathbb{D}\setminus\{0\}$. Finally, $a_{\mathbb{D}}>0$ for $0<\rho<1$, while its declared value at the center is zero. The claimed zero decomposition follows. ☐

If $s = (2/\lambda)\,\mathrm{arctanh}\,\rho$ denotes hyperbolic distance from the center, then

$$a_{\mathbb{D}} = \frac{\mu}{\lambda}\sinh(\lambda s).$$

This identity describes radial propagation on the punctured regular part; it does not repair the failure of differentiability at the center. The metric is regular there, while the assignment is singular.

7.5 Smooth parameter families and their total zero sets

Let $I \subset \mathbb{R}$ be an open interval, let M be a smooth m-manifold without boundary, and let

$$A : M \times I \longrightarrow \mathbb{R}$$

be smooth. Write $a_t(p) = A(p,t)$ and

$$\mathcal{Z} = A^{-1}(0) = \{(p,t) \in M \times I : a_t(p) = 0\}.$$

Proposition 7.7 (Regular total zero set)**.** *Assume that*

$$d_p a_t \neq 0 \qquad \textit{whenever } a_t(p) = 0. \tag{66}$$

Then $\mathcal{Z}$ is a smooth embedded codimension-one submanifold of $M \times I$, and the restriction

$$\pi : \mathcal{Z} \longrightarrow I, \qquad \pi(p,t) = t, \tag{67}$$

is a submersion. In particular, if $\dim M = 2$*, then $\mathcal{Z}$ is a smooth surface.*

Proof. At a point $(p,t) \in \mathcal{Z}$, the total differential is

$$dA_{(p,t)}(v,\tau) = d_p a_t(v) + d_t A_{(p,t)}(\tau).$$

Hypothesis (66) implies that $dA_{(p,t)} \neq 0$, so 0 is a regular value of A. The regular-value theorem proves that $\mathcal{Z}$ is a codimension-one embedded submanifold.

It remains to prove the stronger assertion about the projection. Given an arbitrary $\tau \in T_t I$, put $c = d_t A_{(p,t)}(\tau)$, and choose $w \in T_p M$ with $d_p a_t(w) = 1$, which is possible because $d_p a_t \neq 0$. Set

$$v = -c\,w.$$

Then $dA_{(p,t)}(v,\tau) = 0$, so $(v,\tau) \in T_{(p,t)}\mathcal{Z}$, while $d\pi_{(p,t)}(v,\tau) = \tau$. Hence $d\pi$ is surjective at every point of $\mathcal{Z}$, proving that π is a submersion. □

Remark 7.8 (Properness warning). Proposition 7.7 gives pointwise product charts for the projection, but it does not make $\pi : \mathcal{Z} \to I$ a globally trivial family, or even a fiber bundle over an entire parameter neighborhood when noncompact fibers are present. A standard sufficient hypothesis for local triviality as a smooth fiber bundle is that the projection onto the relevant parameter interval be a proper surjective submersion (with the usual additional care if boundaries are introduced). Without properness, components may escape to or arrive from infinity. Accordingly, $\mathcal{Z}$ is called the *total zero set* here, not a zero tube.

No multi-zero model is included in this foundational chapter. Constructing and classifying such models requires a complete audit of the domain, metric, assignment, singular set, flow equation, and exact zero topology; that problem belongs to the singular-zero theory developed after Paper I.

8 Global Torsion and the Accumulative Commutative Space

The affine cocycle retains the effect of order on evaluation. We now place the same information over a commutative charge plane. This construction has two distinct uses. For a positive add–scale history, it records the total additive and logarithmic multiplicative charges, and it supplies a weighted integral whose value restores the translation part that the charge endpoint alone forgets. The distinction between these two roles is essential: the accumulative commutative space is a shadow of the presented history, not another presentation of its full arithmetic syntax.

Throughout this section an elementary additive step and an elementary scaling step are denoted by

$$\mathsf{A}_p(x) = x + p, \qquad \mathsf{M}_q(x) = e^q x, \qquad p, q \in \mathbb{R}.$$

For a history $\gamma = (g_1, \dots, g_n)$, the entries are performed from left to right, so that

$$\nu_x(\gamma) = g_n \circ \cdots \circ g_1(x).$$

Negative values of p and q are allowed. Thus the same notation covers inverse translations and inverse positive scalings. In this section an *add–scale history* means a finite word in these two families of generators. Such a word evaluates in $\mathrm{Aff}^+(1, \mathbb{R})$. The ACS construction is attached to this presented word; it is not asserted for negative multipliers or for an arbitrary affine syntax before that syntax has been written in add–scale form.

8.1 The charge path

Definition 8.1 (Accumulative commutative space and charge path)**.** The *accumulative commutative space*, abbreviated ACS, is the oriented plane

$$\mathcal{A} = \mathbb{R}^2_{(A,M)}, \qquad \text{with orientation } dA \wedge dM.$$

An additive step A_p has charge increment $(p, 0)$, and a scaling step M_q has charge increment $(0, q)$. Starting at $(A_0, M_0) = (0, 0)$, accumulate these increments in chronological order. The resulting oriented broken path is denoted by C_γ, and its terminal point by

$$(A_\gamma, M_\gamma) = \left(\sum_{g_i = \mathsf{A}_{p_i}} p_i, \sum_{g_i = \mathsf{M}_{q_i}} q_i \right).$$

The terminal charge is insensitive to order, whereas the broken path generally is not. In particular, two histories may have the same ACS endpoint while inducing different affine maps. The missing information is recovered by the following integral formula.

Proposition 8.2 (Direct ACS evaluation)**.** *For every add–scale history γ and every initial value x,*

$$\nu_x(\gamma) = e^{M_\gamma} \left(x + \int_{C_\gamma} e^{-M} \, dA \right). \tag{68}$$

Proof. Let an additive step A_{p_i} occur when the cumulative multiplicative charge is M_{i-1}. Along the corresponding horizontal edge of C_γ,

$$\int e^{-M} \, dA = p_i e^{-M_{i-1}}.$$

After that addition, the remaining multiplicative steps have total charge $M_\gamma - M_{i-1}$. Consequently the contribution of p_i to the final value is

$$p_i e^{M_\gamma - M_{i-1}} = e^{M_\gamma} p_i e^{-M_{i-1}}.$$

The initial value contributes $e^{M_\gamma}x$, and vertical ACS edges contribute nothing to the dA-integral. Summing over the additive steps gives (68). The argument uses oriented edge integrals, so it also covers negative increments. □

The kernel e^{-M} is therefore forced by the affine cocycle: it normalizes an additive contribution to the source frame, before the common terminal scale e^{M_γ} returns it to the target frame.

8.2 Compatibility and relative torsion

There are two useful compatibility conditions, and they should not be conflated. Histories γ and δ are *scale-compatible* if

$$M_\gamma = M_\delta.$$

They are *charge-compatible* if, in addition,

$$A_\gamma = A_\delta.$$

Scale compatibility makes their affine maps have the same linear part. Charge compatibility makes their ACS paths have both endpoints in common and hence makes $C_\gamma - C_\delta$ a closed oriented one-chain.

Definition 8.3 (Relative arithmetic torsion)**.** For scale-compatible add–scale histories γ and δ, their *relative arithmetic torsion* is

$$\tau(\gamma, \delta) := \xi_\gamma - \xi_\delta, \tag{69}$$

where ξ_γ and ξ_δ are the target-frame translation terms of the two affine operators. This relative translation defect is the primary torsion invariant used below.

Proposition 8.4. *For every initial value x, relative arithmetic torsion equals the endpoint difference*

$$\tau(\gamma, \delta) = \nu_x(\gamma) - \nu_x(\delta),$$

and is therefore independent of x. If $M_\gamma = M_\delta = M_*$, then

$$\tau(\gamma, \delta) = e^{M_*} \left(\int_{C_\gamma} e^{-M}\, dA - \int_{C_\delta} e^{-M}\, dA \right). \tag{70}$$

Proof. Apply Proposition 8.2 to both histories. Their initial terms are respectively $e^{M_\gamma}x$ and $e^{M_\delta}x$, which cancel under scale compatibility. The remaining terms give (70). □

Temporal reversal gives one possible comparison, but it is not built into the definition. The pair (γ, δ) may consist of any two scale-compatible add–scale histories. This extension is important because order defects occur between histories that are neither reversals nor planar mirrors of one another.

8.3 The torsion–Stokes formula

Suppose now that γ and δ are charge-compatible, with common terminal charge (A_*, M_*). Define the target-normalized one-form

$$\eta_* := e^{M_* - M}\, dA \tag{71}$$

on the ACS. Relative torsion is then the integral of η_* over the closed one-chain $C_\gamma - C_\delta$. The chosen order of the two paths fixes the sign.

Theorem 8.5 (Weighted torsion–Stokes theorem)**.** *Let γ and δ be charge-compatible add–scale histories with common terminal multiplicative charge M_*. Let $\Sigma_{\gamma,\delta}$ be any oriented piecewise smooth singular two-chain in the ACS satisfying*

$$\partial\Sigma_{\gamma,\delta} = C_\gamma - C_\delta.$$

Then

$$\begin{aligned} \tau(\gamma,\delta) &= \int_{C_\gamma - C_\delta} \eta_* && (72)\\ &= \int_{\Sigma_{\gamma,\delta}} e^{M_*-M}\, dA \wedge dM. && (73) \end{aligned}$$

The value of the area integral is independent of the chosen filling chain.

Proof. Equation (70) and charge compatibility give

$$\tau(\gamma,\delta) = \int_{C_\gamma - C_\delta} e^{M_*-M}\, dA = \int_{C_\gamma - C_\delta} \eta_*.$$

With the orientation $dA \wedge dM$, direct differentiation yields

$$d\eta_* = d(e^{M_*-M}) \wedge dA = -e^{M_*-M} dM \wedge dA = e^{M_*-M} dA \wedge dM.$$

Stokes' theorem for piecewise smooth chains therefore gives

$$\int_{C_\gamma - C_\delta} \eta_* = \int_{\partial\Sigma_{\gamma,\delta}} \eta_* = \int_{\Sigma_{\gamma,\delta}} d\eta_*,$$

which is (73). Such a filling exists because the ACS plane is contractible. Finally, if Σ and Σ' have the same boundary, applying Stokes' theorem to each shows

$$\int_\Sigma d\eta_* = \int_{\partial\Sigma} \eta_* = \int_{\partial\Sigma'} \eta_* = \int_{\Sigma'} d\eta_*.$$

This also proves filling independence. The chain formulation remains valid when the broken paths self-intersect or carry signed increments; no informal notion of an enclosed region is required. □

Charge compatibility is needed for the closed-chain and area formulas, but not for Definition 8.3. A merely scale-compatible pair still has a well-defined translation defect, even when the endpoints of its ACS paths have different A-coordinates.

Figure 7 depicts the chain orientation used in Theorem 8.5. It is a schematic of the theorem's data rather than an argument for filling independence.

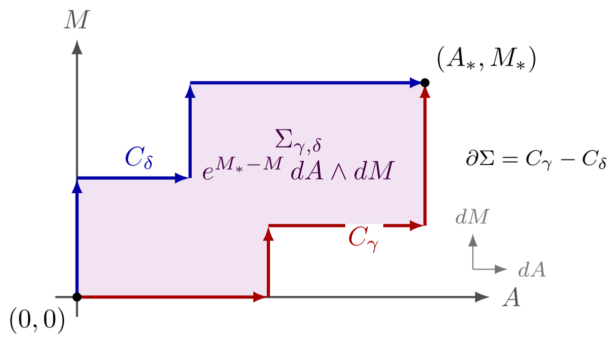

Figure 7: Two charge-compatible ACS histories. Both paths run from $(0,0)$ to the common charge endpoint (A_*, M_*). Traversing C_γ forward and C_δ backward gives the positive boundary of the displayed filling for the orientation $dA \wedge dM$; the density is weighted by e^{M_*-M}.

8.4 Explicit order comparisons

The elementary rectangle. Let

$$\gamma = (\mathsf{A}_p, \mathsf{M}_q), \qquad \delta = (\mathsf{M}_q, \mathsf{A}_p).$$

The path $C_\gamma - C_\delta$ is the positively oriented boundary of the rectangle $[0,p] \times [0,q]$ when $p, q > 0$. Direct evaluation and the weighted-area formula agree:

$$\begin{aligned}\tau(\gamma,\delta) &= e^q(x+p) - (e^q x + p)\\ &= p(e^q - 1)\\ &= \int_0^p \int_0^q e^{q-M}\, dM\, dA.\end{aligned}$$

Thus "add, then multiply" minus "multiply, then add" is positive under the orientation fixed in Definition 8.1. In the flow notation used in the next chapter, $p = \mu h$ and $q = \lambda k$, so the defect is

$$\mu h(e^{\lambda k} - 1).$$

A four-step staircase. For

$$\gamma = (\mathsf{A}_p, \mathsf{M}_r, \mathsf{A}_q, \mathsf{M}_s), \qquad \delta = (\mathsf{M}_r, \mathsf{A}_p, \mathsf{M}_s, \mathsf{A}_q),$$

the histories have the same total charge $(p+q, r+s)$, but δ is not the temporal reversal of γ. Their evaluations are

$$\nu_x(\gamma) = e^{r+s}x + pe^{r+s} + qe^s,$$
$$\nu_x(\delta) = e^{r+s}x + pe^s + q.$$

Consequently

$$\tau(\gamma,\delta) = pe^s(e^r - 1) + q(e^s - 1). \tag{74}$$

For positive p, q, r, s, both terms are positive weighted strips in an ACS filling. Formula (74) also shows how a later scale changes the weight of an earlier commutation defect.

Equal charges without reversal. The pair

$$(\mathsf{A}_p, \mathsf{M}_r, \mathsf{A}_q), \qquad (\mathsf{A}_q, \mathsf{M}_r, \mathsf{A}_p)$$

has common charge $(p+q, r)$, and its relative torsion is

$$(p-q)(e^r - 1).$$

It may be positive, negative, or zero. In particular, equal charge does not determine torsion, while zero torsion does not imply equality of marked histories.

8.5 What the ACS forgets

At the formal level, the map

$$\gamma \longmapsto (A_\gamma, M_\gamma)$$

is an order-forgetting charge homomorphism from concatenated marked histories to the additive group $\mathbb{R}^2$. It is therefore analogous to an abelianization map. This statement concerns the formal charge language only. It does not identify the symbolic history space with the evaluated affine group, and it does not assert that either object is a free group without an additional choice of formal generators and relations.

The content of Theorem 8.5 is precisely what survives this loss of order: the endpoint charges specify a common frame, while the weighted filling measures the translation defect between two histories. The next chapter retains the current value as a third coordinate and identifies the infinitesimal form of the same defect with contact curvature.

9 Contact Connection and Horizontal Curvature

The ACS records the ordered add–scale charge path needed for evaluation while forgetting the current assignment value and some syntactic provenance. A local model should retain that value and should distinguish the infinitesimal additive and multiplicative directions. The resulting three-dimensional state space carries both an Ehresmann connection over the two-dimensional direction plane and, in the nondegenerate case, a contact form.

Fix real constants μ and λ, and let

$$\mathcal{C} = \mathbb{R}^3_{(u,v,a)}, \qquad \pi : \mathcal{C} \longrightarrow \mathbb{R}^2_{(u,v)}.$$

Here a is the current assignment value, while u and v parametrize additive and multiplicative motion. We orient $\mathcal{C}$ by

$$du \wedge da \wedge dv.$$

9.1 The arithmetic contact form

Define

$$\alpha := da - \mu\, du - \lambda a\, dv. \tag{75}$$

The equation $\alpha = 0$ says that a lifted base motion obeys the affine propagation law.

Proposition 9.1 (Contact nondegeneracy)**.** *With the orientation $du \wedge da \wedge dv$,*

$$\alpha \wedge d\alpha = \mu\lambda\, du \wedge da \wedge dv. \tag{76}$$

Hence α is a contact form if and only if $\mu\lambda \neq 0$.

Proof. Since μ and λ are constant,

$$d\alpha = -\lambda\, da \wedge dv.$$

It follows that

$$\begin{aligned} \alpha \wedge d\alpha &= (da - \mu\, du - \lambda a\, dv) \wedge (-\lambda\, da \wedge dv) \\ &= \mu\lambda\, du \wedge da \wedge dv. \end{aligned}$$

This three-form is nowhere zero exactly when $\mu\lambda \neq 0$. □

The contact-isomorphism class is the standard local one; the arithmetic content lies in the coordinates and in the preferred horizontal frame introduced next. Even in a degenerate case, $\ker \alpha$ remains a well-defined horizontal distribution, although it is then not contact.

9.2 Horizontal lifts and the affine flow

Set

$$D_u := \partial_u + \mu\partial_a, \qquad D_v := \partial_v + \lambda a\partial_a. \tag{77}$$

Then

$$\alpha(D_u) = \alpha(D_v) = 0,$$

and the two fields project respectively to ∂_u and ∂_v. Thus

$$\mathcal{D} := \ker \alpha = \operatorname{span}\{D_u, D_v\}$$

is an Ehresmann connection for π. Its additive and multiplicative flows are

$$\Phi_u^h(u, v, a) = (u + h, v, a + \mu h), \tag{78}$$
$$\Phi_v^k(u, v, a) = (u, v + k, e^{\lambda k} a). \tag{79}$$

For a direction angle θ, define

$$D_\theta = \cos\theta\, D_u + \sin\theta\, D_v.$$

If $(u(s), v(s), a(s))$ is tangent to D_θ, then

$$\frac{du}{ds} = \cos\theta, \qquad \frac{dv}{ds} = \sin\theta,$$

and

$$\frac{da}{ds} = \mu\cos\theta + \lambda a \sin\theta. \tag{80}$$

Thus the horizontal lift reproduces the continuous affine flow rather than postulating a separate dynamics.

9.3 Horizontal curvature

Theorem 9.2 (Contact curvature)**.** *The preferred horizontal fields satisfy*

$$[D_u, D_v] = \mu\lambda\partial_a. \tag{81}$$

In particular, when $\mu\lambda \neq 0$, the horizontal distribution is not integrable. Its bracket defect is vertical and has constant coefficient $\mu\lambda$ in the arithmetic frame.

Proof. Using (77),

$$\begin{aligned} [D_u, D_v] &= [\partial_u + \mu\partial_a, \partial_v + \lambda a\partial_a] \\ &= \mu\,\partial_a(\lambda a)\partial_a - \lambda a\,\partial_a(\mu)\partial_a \\ &= \mu\lambda\partial_a. \end{aligned}$$

All remaining coordinate brackets vanish. Since ∂_a is vertical and is not contained in $\ker\alpha$, the bracket is the vertical failure of horizontal closure. □

The sign in (81) agrees with the chronological convention of the preceding chapter. Equivalently,

$$\alpha([D_u, D_v]) = \mu\lambda, \qquad d\alpha(D_u, D_v) = -\mu\lambda.$$

The two formulas have opposite signs by the identity $d\alpha(X, Y) = -\alpha([X, Y])$ for horizontal fields X, Y.

9.4 Open defects and closed horizontal loops

The bracket is infinitesimal. At finite scale there are two related, but different, quantities. First compare the two open lifts from (u, v) to $(u + h, v + k)$:

$$\begin{aligned} \Phi_v^k \circ \Phi_u^h(u, v, a) &= (u + h, v + k, e^{\lambda k}(a + \mu h)), \\ \Phi_u^h \circ \Phi_v^k(u, v, a) &= (u + h, v + k, e^{\lambda k}a + \mu h). \end{aligned}$$

Their assignment difference, with "add then multiply" taken first, is

$$T_{\text{open}}(h, k) = \mu h(e^{\lambda k} - 1). \tag{82}$$

This is exactly the elementary ACS rectangle torsion with charge increments $(\mu h, \lambda k)$.

Now perform the chronological four-step history

$$\mathsf{A}_h, \quad \mathsf{M}_k, \quad \mathsf{A}_{-h}, \quad \mathsf{M}_{-k},$$

where A_h denotes the flow Φ_u^h and M_k denotes Φ_v^k. The base point returns to (u, v), and direct substitution gives

$$\begin{aligned} a &\longmapsto a + \mu h \longmapsto e^{\lambda k}(a + \mu h) \\ &\longmapsto e^{\lambda k}(a + \mu h) - \mu h \longmapsto a + \mu h(1 - e^{-\lambda k}). \end{aligned}$$

The closed-loop vertical drift is therefore

$$T_{\text{closed}}(h, k) = \mu h(1 - e^{-\lambda k}) = e^{-\lambda k} T_{\text{open}}(h, k). \tag{83}$$

Thus the finite open comparison and the finite closed holonomy are not equal. Here T_{closed} is the source-normalized relative translation. Converting it to the target frame gives $e^{\lambda k} T_{\text{closed}} = T_{\text{open}}$ exactly.

Figure 8 keeps the two finite constructions visually separate. The square is drawn in the (u, v) base; the short vertical segments represent assignment fibers, not additional base edges.

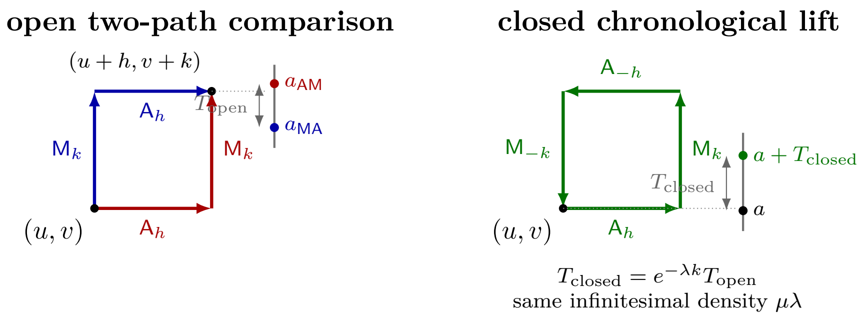

Figure 8: Open defect and closed holonomy in the contact model. The two open horizontal lifts have the same base endpoint but different assignment values. The four-step closed base loop returns to its starting base point with a raw vertical drift. These finite quantities differ by the displayed scale factor; converting the closed drift to the target frame gives the open defect, while their infinitesimal densities agree as stated in the text.

9.5 The scalar horizontal differential

For a smooth scalar field $F \in C^\infty(\mathcal{C})$, define

$$\delta_H F := dF - (\partial_a F)\alpha = (D_u F)\, du + (D_v F)\, dv. \tag{84}$$

This is the ordinary differential restricted to horizontal directions and written on the base coframe. In particular,

$$\delta_H a = \mu\, du + \lambda a\, dv.$$

For scalar fields only, define the antisymmetrized second horizontal derivative by

$$\mathcal{R}_H(F) := (D_u D_v F - D_v D_u F)\, du \wedge dv. \tag{85}$$

This notation deliberately avoids presupposing an extension of δ_H to a graded complex.

Proposition 9.3 (Curvature on scalar fields)**.** *For every $F \in C^\infty(\mathcal{C})$,*

$$\mathcal{R}_H(F) = \mu\lambda(\partial_a F)\, du \wedge dv. \tag{86}$$

In particular,

$$\mathcal{R}_H(a) = \mu\lambda\, du \wedge dv.$$

Proof. By definition and Theorem 9.2,

$$\mathcal{R}_H(F) = [D_u, D_v](F)\, du \wedge dv = \mu\lambda(\partial_a F)\, du \wedge dv.$$

Taking $F = a$ proves the final formula. □

One may use $\delta_H^2 F$ as shorthand for $\mathcal{R}_H(F)$ after declaring this antisymmetrization convention. It does not mean that δ_H has been defined on horizontal forms, nor does it assert the existence of a nilpotent differential complex. No such graded calculus is required for the foundational result.

9.6 Three levels of the same order defect

Proposition 9.4 (Finite and infinitesimal comparison)**.** *For the elementary rectangle with additive side μh and logarithmic multiplicative side λk, the following statements hold.*

1. *The open affine defect equals the ACS weighted area exactly:*

$$T_{\text{open}}(h,k) = \int_{[0,\mu h]\times[0,\lambda k]} e^{\lambda k - M}\, dA \wedge dM = \mu h(e^{\lambda k} - 1),$$

   with oriented-chain interpretation when a side is negative.

2. *The closed chronological lift has exact drift*

$$T_{\text{closed}}(h,k) = e^{-\lambda k} T_{\text{open}}(h,k).$$

   Thus T_{closed} is the source-normalized relative translation; its target-frame conversion agrees with the open defect, but the raw finite values generally differ.

3. *Both defects have the common infinitesimal density*

$$\lim_{(h,k)\to(0,0)} \frac{T_{\text{open}}(h,k)}{hk} = \lim_{(h,k)\to(0,0)} \frac{T_{\text{closed}}(h,k)}{hk} = \mu\lambda,$$

   where the quotients are taken for $hk \neq 0$. This is the coefficient of $[D_u, D_v]$ and of $\mathcal{R}_H(a)$.

Proof. The first identity is Theorem 8.5 applied to the elementary rectangle, and its integral evaluates to

$$\mu h \int_0^{\lambda k} e^{\lambda k - M}\, dM = \mu h(e^{\lambda k} - 1).$$

The second identity is (83). Finally,

$$\frac{e^{\lambda k} - 1}{k} \longrightarrow \lambda, \qquad \frac{1 - e^{-\lambda k}}{k} \longrightarrow \lambda,$$

which proves the two limits. The curvature identifications follow from Theorem 9.2 and Proposition 9.3. $\square$

The synthesis has three layers, not one finite equality: exact target-frame open affine torsion is measured by ACS weighted area; source-normalized closed horizontal holonomy differs by an exact frame-conversion factor; and contact curvature is their common infinitesimal limit. The contact distribution alone supplies neither a preferred complex structure nor a full analytic function theory. The present paper uses only the horizontal connection, its scalar differential, and its curvature.

10 Conclusions and Research Interfaces

This paper has established a foundational passage from sequential arithmetic syntax to geometry. The organizing principle is that an expression should not be identified prematurely with its endpoint value. An unmarked sequential tree requires a choice of accumulator; the resulting fully marked planar tree is equivalent to a bounded marked history. From that history, operator evaluation and the ACS charge shadow form separate branches, while an initial value turns an operator into an endpoint. The non-bijective forgetful maps lose information, and several of the structures developed here measure precisely what is lost.

10.1 Results of the foundational construction

The first part of the paper classified binary expression trees with a unique legal internal evaluation order and represented them by marked spinal histories. Operand-slot data made it possible to distinguish planar mirror, temporal reversal, and path inverse. These operations act at different levels and need not induce the same operator comparison.

For nondegenerate bilateral one-hole contexts, projective continuation produced Möbius transformations. Under the algebraic hypotheses stated there, the elementary contexts generate PGL_2, while ordinary addition and nonzero scaling form the affine, or Borel, sector stabilizing the chosen point at infinity. This placement is important: projective semantics extends the operator domain, but it does not turn a pole into an admissible ordinary arithmetic operation.

The smaller alphabet consisting of translation by $\pm\sqrt{2}$ and negative inversion realizes the $q = 4$ Hecke group inside this projective semantics. Its matrix relation $(JT)^4 = 1$ gives an explicit pair of different marked histories with the same operator. The subsequent factorization through cell stabilizers then separates four levels that are needed downstream: history, operator, cell coset, and endpoint. In particular, the arithmetic words are not identified one-to-one with tiles in a chosen Hecke cellulation.

Within the affine sector, chronological composition yielded explicit cocycle coordinates and the continuous equation

$$\frac{da}{ds} = \mu \cos\theta + \lambda a \sin\theta.$$

The basic upper-half-plane model realizes this equation geometrically and gives a first regular arithmetic expression space. Separately, the regular-value argument shows that a zero in a regular AES lies on a smooth codimension-one zero set. Isolated or singular zeros must therefore arise from a failure of those regular hypotheses, not from the regular theory itself. Under the additional assumptions that the metric is complete and the manifold is connected and boundaryless, the unit-gradient rectification gives the stronger diffeomorphism

$$M \cong Z(a) \times \mathbb{R}.$$

Hence the zero layer is nonempty and connected. This is not asserted for incomplete models or manifolds with boundary, and it is not in general a Riemannian product decomposition.

The final two chapters identified global and local forms of arithmetic order defect. In the accumulative commutative space, a positive add–scale history γ satisfies

$$\nu_x(\gamma) = e^{M_\gamma}\left(x + \int_{C_\gamma} e^{-M}\, dA\right),$$

and any charge-compatible pair of such histories has relative torsion equal to the integral of the weighted area form over an oriented filling chain. On the local state space, the horizontal fields associated with

$$\alpha = da - \mu\, du - \lambda a\, dv$$

satisfy

$$[D_u, D_v] = \mu\lambda\partial_a.$$

The exact target-frame open defect, the source-normalized closed-loop drift, and the infinitesimal curvature were kept distinct. Converting the closed drift back to the target frame gives the open defect; the two raw defects share only the stated infinitesimal limit.

The proved logical spine may therefore be summarized as

$$\begin{gathered}\text{marked planar sequential tree} \longleftrightarrow \text{bounded marked history} \longrightarrow \text{projective operator},\\ \mathrm{PGL}_2 \supset \mathrm{Aff}(1) \longrightarrow \text{affine cocycle and flow},\\ \mathrm{Hist}_4 \longrightarrow G_4 \longrightarrow G_4/H_{\mathcal{C}} \longrightarrow \text{Hecke cell},\\ \text{complete boundaryless regular AES} \longrightarrow Z(a) \times \mathbb{R},\\ \text{charge-compatible positive add–scale defect} \longrightarrow \text{ACS weighted area},\\ \text{infinitesimal defect} \longrightarrow \text{contact curvature}.\end{gathered}$$

This is a foundation for later developments, not a claim that every arithmetic expression, singularity, or computational phenomenon is already captured by the affine theory.

10.2 Interface to Paper II

Arithmetic Expression Geometry II: Hyperbolic Real Function Theory will begin where the present horizontal connection stops. Its program is to choose and analyze additional metric and analytic data: a horizontal metric, compatible complex or almost-complex structures where appropriate, second-order operators, boundary-value constructions, and arithmetic holomorphicity. None of these structures follows uniquely from the contact form alone, and no result of that prospective function theory is used in the proofs of Paper I. The G_4 sublanguage supplies an additional, explicitly arithmetic test case: an automorphic or equivariant function used there must be derived from the group action and must keep its analytic descent conditions separate from ordinary arithmetic admissibility of a history at a real input.

10.3 Interface to Paper III

Arithmetic Expression Geometry III: Singular Zero Geometry and Tubes studies the boundary of the regular-zero theorem. Its objects include singular and multiple zeros, parameter discriminants, regular and singular zero surfaces, and tube constructions. Braid, monodromy, and knot structures there require separate invariance and normalization arguments. The Hecke interface proved here offers a source-driven alternative to imposing a four-valent cellulation by hand: Paper III uses a normalized Hauptmodul to construct the corresponding singular zero dessin, then relates its sign cover to Hecke periodic-orbit knots. Those analytic, singular, and three-dimensional results remain downstream results, not results of Paper I. The complete regular splitting theorem shows that such a branching network cannot be the zero set of one connected, complete, boundaryless regular scalar AES; any singularity, boundary, or incompleteness used in the construction must therefore be stated explicitly. In particular, the existence of a tube does not by itself produce a knot invariant, and no such general invariant is asserted here.

10.4 Interface to Paper IV

Arithmetic Expression Geometry IV: Projective Condensation and Computational Complexity will return to the full projective process–result hierarchy. Its proposed scope includes bivaluation, quotient and condensation constructions, representation complexity, and the relation between geometric and computational resources. The affine embedding in projective semantics proved in this paper supplies an entry point, but it does not establish any general complexity theorem. Those claims remain open until precise models of representation, input size, and computation are fixed.

10.5 Open foundational problems

Several questions remain at the level of foundations.

1. Classify regular affine arithmetic expression spaces beyond the basic model on the upper half-plane, including the hypotheses under which two models should count as equivalent.
2. Construct a genuinely projective replacement for the affine ACS and determine which part of the weighted torsion formula survives without a preferred point at infinity.
3. Relate affine contact curvature to projective holonomy without conflating an infinitesimal bracket with a finite transport operator.
4. Characterize the marked-history fibers of the downstream G_4 operator–zero-network realization and determine what arithmetic decoration, if any, survives its operator and knot quotients.
5. Produce fully verified singular multi-zero models and identify their stable local normal forms and discriminants.
6. Formulate, and then test, a quantitative relation between geometric history complexity, representation complexity, and computational complexity.

These are research interfaces rather than consequences of the present paper. The stable conclusion is narrower and more concrete: sequential arithmetic histories possess projective operator semantics; their affine sector carries a natural cocycle and flow; and the failure of additive and multiplicative order to commute admits exact global and infinitesimal geometric descriptions.

A Composition, action, and domain conventions

This appendix collects the convention-sensitive choices used in Sections 2–4. The choices are not mathematical assertions, but every composition, sign, and coordinate formula depends on them.

A.1 Chronological words and functional composition

Convention A.1 (Order inside a history)**.** A history

$$\gamma = (g_1, \ldots, g_n)$$

performs g_1 first and g_n last. Its functional evaluation is therefore

$$\mathrm{ev}(\gamma) = g_n \circ \cdots \circ g_1$$

when the g_i are viewed as maps.

Convention A.2 (Chronological concatenation)**.** The written product $\gamma\delta$ means first perform γ and then perform δ. Thus

$$\mathrm{ev}(\gamma\delta) = \mathrm{ev}(\delta) \circ \mathrm{ev}(\gamma).$$

The notation is chronological rather than functional: the earlier process is written on the left.

For example, if $g(z) = z + \mu$ and $h(z) = qz$, then the word $\gamma = (g, h)$ and the chronological product of the one-step histories gh both act as

$$h \circ g(z) = qz + q\mu.$$

The reverse chronological word hg acts as $g \circ h(z) = qz + \mu$. This two-step check fixes the sign of their endpoint difference:

$$\mathrm{ev}(gh)(z) - \mathrm{ev}(hg)(z) = \mu(q - 1).$$

A.2 Column-vector projective action

Convention A.3 (Matrix action)**.** Homogeneous coordinates are columns, and matrices act from the left:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} Au + Bv \\ Cu + Dv \end{pmatrix}.$$

On the affine chart $z = [z : 1]$, this gives

$$M \cdot z = \frac{Az + B}{Cz + D}.$$

If M_i represents g_i, the history $(g_1, \ldots, g_n)$ is represented by

$$M_n \cdots M_1.$$

Consequently, for chronological concatenation,

$$\rho(\gamma\delta) = \rho(\delta)\rho(\gamma). \tag{87}$$

This is the matrix version of Convention A.2. A row-vector right action would require a different product convention; it is not used in this paper.

Projective matrices are scalar classes. If M and aM differ by $a \in K^\times$, they represent the same element of $\mathrm{PGL}_2(K)$. Matrix identities used for projective maps may therefore be checked either exactly in $\mathrm{GL}_2(K)$ or up to one overall nonzero scalar, provided the choice is stated.

A.3 Ordinary and projective domains

The following table records the domain distinction for the contexts involving division. Here $c \in K$ and z is the evolving affine value.

Context	Ordinary rule	Ordinary condition	Projective status
$C^{(1)}_{/,c}[z] = z/c$	divide by fixed c	$c \neq 0$	invertible if $c \neq 0$
$C^{(2)}_{/,c}[z] = c/z$	divide by evolving z	$z \neq 0$	invertible if $c \neq 0$
$C^{(\varepsilon)}_{\times,c}[z] = cz$	multiply by c	none	invertible if $c \neq 0$

Two different exclusions appear in the table. The parameter condition $c \neq 0$ ensures that the context has an invertible projective matrix. The state condition $z \neq 0$ ensures that an ordinary second-slot division is defined at that step. Projectively, a non-degenerate c/z sends 0 to ∞; this is a chart transition, not an ordinary quotient in K.

For a composite history, ordinary admissibility is checked at every intermediate state. It is not enough that the final projective point lies in the affine chart. In particular, two projective inversions compose to the identity operator, but the corresponding ordinary two-step word is not admissible at zero.

A.4 Left and right group data

The affine group is represented by

$$g(\Phi, \xi) = \begin{pmatrix} \Phi & \xi \\ 0 & 1 \end{pmatrix}, \qquad \Phi \in K^\times.$$

With the column action, matrix multiplication gives

$$g(\Phi_2, \xi_2)g(\Phi_1, \xi_1) = g(\Phi_2\Phi_1, \Phi_2\xi_1 + \xi_2). \tag{88}$$

Thus the earlier translation is rescaled by the later affine map. Equation (88) is the two-factor origin of the target-frame cocycle.

Over the positive real component, write $\Phi = e^M$. The two Maurer–Cartan forms used in Section 4 have the following readings.

Form	Translation component	Frame
$g^{-1}dg$	$e^{-M}d\xi$	body/source
$dg\, g^{-1}$	$d\xi - \xi dM$	spatial/target

The words "left" and "right" here refer to invariant differential forms on the affine group. They do not refer to first and second operand slots. The operand-slot mirror is syntactic; the Maurer–Cartan distinction is group-theoretic.

A.5 Target and source-normalized translation coordinates

For $x \mapsto \Phi x + \xi$, the symbol ξ always denotes target-frame translation. The source-normalized coordinate is

$$\widehat{\xi} = \Phi^{-1}\xi.$$

Their discrete and differential conversions are

$$\xi = \Phi\widehat{\xi}, \qquad \widehat{\xi} = \Phi^{-1}\xi,$$

and, when $\Phi = e^M$,

$$d\xi = \xi\, dM + e^M d\widehat{\xi}.$$

No formula in the main text changes from ξ to $\widehat{\xi}$ without this normalization being displayed.

A.6 Mirror, reversal, and inverse notation

For $\gamma = (g_1, \ldots, g_n)$, the three operations are summarized by

Operation	Resulting word	Required hypothesis
Mirror $m\gamma$	$(m(g_1), \ldots, m(g_n))$	none
Temporal reversal $r\gamma$	$(g_n, \ldots, g_1)$	none
Path inverse γ^{-1}	$(g_n^{-1}, \ldots, g_1^{-1})$	every g_i invertible

Mirror exchanges operand slots and retains time. Temporal reversal changes time and retains the original contexts. Path inverse reverses time and inverts every operator. Accordingly,

$$\rho(\gamma^{-1}) = \rho(\gamma)^{-1},$$

whereas no analogous identity holds for $r\gamma$ in general.

A.7 Notation conversion summary

The table below lists older or potentially ambiguous descriptions and the notation used in Paper I.

Potentially ambiguous description	Paper I notation
evolving value in operand slot one	$C^{(1)}_{\omega,c}[z] = \omega(z,c)$
evolving value in operand slot two	$C^{(2)}_{\omega,c}[z] = \omega(c,z)$
chronological context word	$\gamma = (g_1, \dots, g_n)$, g_1 first
history product	$\gamma\delta$, first γ, then δ
operator evaluation	$\rho(\gamma)$
ordinary endpoint at x	$\nu_x(\gamma)$
accumulated scale	Φ_γ
target-frame translation	ξ_γ
source-normalized translation	$\widehat{\xi}_\gamma = \Phi_\gamma^{-1}\xi_\gamma$
operand-slot exchange	$m\gamma$
time-order reversal	$r\gamma$
inverse operator path	γ^{-1}

B Affine cocycle calculations

This appendix supplements Proposition 4.1 with expanded calculations. All discrete formulas hold over an arbitrary field K, and every multiplier is assumed nonzero.

B.1 Prefix recursion and closed form

For the prefix history $(f_1, \dots, f_k)$, write

$$F_k = f_k \circ \cdots \circ f_1, \qquad F_k(x) = \Phi_k x + \xi_k,$$

with $F_0(x) = x$. If $f_{k+1}(x) = s_{k+1}x + t_{k+1}$, then

$$\Phi_{k+1} = s_{k+1}\Phi_k, \qquad \xi_{k+1} = s_{k+1}\xi_k + t_{k+1}. \tag{89}$$

Iterating the first recursion gives

$$\Phi_k = s_k \cdots s_1.$$

Iterating the second one gives

$$\begin{aligned}
\xi_1 &= t_1, \\
\xi_2 &= s_2 t_1 + t_2, \\
\xi_3 &= s_3 s_2 t_1 + s_3 t_2 + t_3, \\
\xi_4 &= s_4 s_3 s_2 t_1 + s_4 s_3 t_2 + s_4 t_3 + t_4.
\end{aligned}$$

The pattern is

$$\xi_k = \sum_{i=1}^{k} t_i \prod_{j=i+1}^{k} s_j.$$

Substitution in (89) proves this formula inductively, including the final term through the empty-product convention.

The source-normalized prefix coordinate $\widehat{\xi}_k = \Phi_k^{-1}\xi_k$ obeys a different recursion. From (89),

$$\widehat{\xi}_{k+1} = \frac{s_{k+1}\xi_k + t_{k+1}}{s_{k+1}\Phi_k} = \widehat{\xi}_k + \frac{t_{k+1}}{\Phi_{k+1}}. \tag{90}$$

Starting from $\widehat{\xi}_0 = 0$ therefore gives

$$\widehat{\xi}_k = \sum_{i=1}^{k} \frac{t_i}{\Phi_i} = \sum_{i=1}^{k} \frac{t_i}{\prod_{j=1}^{i} s_j}.$$

This second derivation makes the past normalization explicit at every prefix.

B.2 Suffix transport and the future-weight kernel

For $0 \le i \le n$, define the scale of the strict suffix after step i by

$$S_{i\to n} = \prod_{j=i+1}^{n} s_j, \qquad S_{n\to n} = 1.$$

Then the target-frame translation is simply

$$\xi_n = \sum_{i=1}^{n} S_{i\to n} t_i. \tag{91}$$

To verify the interpretation, replace only t_i by $t_i + \epsilon$. All states before step i are unchanged. At step i the output gains ϵ, and every later affine map multiplies this change by its scale. Thus the final endpoint changes by

$$\epsilon s_{i+1} \cdots s_n = \epsilon S_{i\to n}.$$

The coefficient in (91) is therefore exactly the transport of an additive event to the target frame.

B.3 Alternating addition–scale histories

Consider a positive real alternating history with pairs

$$x \xmapsto{e^{m_i}} e^{m_i}x \xmapsto{+\mu_i} e^{m_i}x + \mu_i, \qquad i = 1, \ldots, n,$$

applied from $i = 1$ to $i = n$. At the level of affine steps this has $s_i = e^{m_i}$ and $t_i = \mu_i$. Define prefix and suffix logarithmic scales by

$$M_i = \sum_{j=1}^{i} m_j, \qquad M_{i+1:n} = \sum_{j=i+1}^{n} m_j.$$

Proposition 4.1 becomes

$$\Phi_n = e^{M_n}, \tag{92}$$

$$\xi_n = \sum_{i=1}^{n} \mu_i e^{M_{i+1:n}}, \tag{93}$$

$$\widehat{\xi}_n = \sum_{i=1}^{n} \mu_i e^{-M_i}. \tag{94}$$

For $n = 3$, direct expansion checks the target formula:

$$\begin{aligned} x &\longmapsto e^{m_1}x + \mu_1 \\ &\longmapsto e^{m_2+m_1}x + e^{m_2}\mu_1 + \mu_2 \\ &\longmapsto e^{m_3+m_2+m_1}x + e^{m_3+m_2}\mu_1 + e^{m_3}\mu_2 + \mu_3. \end{aligned}$$

Dividing the translation line by $e^{m_1+m_2+m_3}$ gives

$$\widehat{\xi}_3 = \mu_1 e^{-m_1} + \mu_2 e^{-(m_1+m_2)} + \mu_3 e^{-(m_1+m_2+m_3)},$$

in agreement with (94).

The convention in this example is "scale, then add" within each pair. If a source instead groups the elementary operations as "add, then scale", the translation attached to that pair is $e^{m_i}\mu_i$, not μ_i. Converting the pair to the common affine form $s_i x + t_i$ before using the cocycle formulas prevents this convention change from being hidden.

B.4 Concatenation in target and normalized coordinates

Let

$$\rho(\gamma) = g(\Phi_\gamma, \xi_\gamma), \qquad \rho(\delta) = g(\Phi_\delta, \xi_\delta),$$

where $g(\Phi, \xi)(x) = \Phi x + \xi$. Since $\gamma\delta$ performs γ first,

$$\begin{aligned} \rho(\gamma\delta)(x) &= \rho(\delta)(\rho(\gamma)(x)) \\ &= \Phi_\delta(\Phi_\gamma x + \xi_\gamma) + \xi_\delta \\ &= \Phi_\delta\Phi_\gamma x + \Phi_\delta\xi_\gamma + \xi_\delta. \end{aligned}$$

This yields the target-coordinate pair

$$(\Phi_{\gamma\delta}, \xi_{\gamma\delta}) = (\Phi_\delta\Phi_\gamma, \Phi_\delta\xi_\gamma + \xi_\delta).$$

For normalized coordinates,

$$\begin{aligned} \widehat{\xi}_{\gamma\delta} &= \frac{\Phi_\delta\xi_\gamma + \xi_\delta}{\Phi_\delta\Phi_\gamma} \\ &= \widehat{\xi}_\gamma + \Phi_\gamma^{-1}\widehat{\xi}_\delta. \end{aligned}$$

The first formula transports the earlier translation forward by the later scale. The second transports the later normalized translation back through the earlier scale.

For three histories, repeated use gives

$$\begin{aligned} \xi_{\gamma\delta\epsilon} &= \Phi_\epsilon\Phi_\delta\xi_\gamma + \Phi_\epsilon\xi_\delta + \xi_\epsilon, \\ \widehat{\xi}_{\gamma\delta\epsilon} &= \widehat{\xi}_\gamma + \Phi_\gamma^{-1}\widehat{\xi}_\delta + (\Phi_\delta\Phi_\gamma)^{-1}\widehat{\xi}_\epsilon. \end{aligned}$$

These are the block-history versions of future weighting and past normalization.

B.5 Inverse histories and relative comparison

The affine inverse is

$$g(\Phi,\xi)^{-1} = g(\Phi^{-1}, -\Phi^{-1}\xi). \tag{95}$$

Indeed,

$$g(\Phi^{-1}, -\Phi^{-1}\xi)(\Phi x + \xi) = x + \Phi^{-1}\xi - \Phi^{-1}\xi = x.$$

Using (95), the relative affine defect is

$$\rho(\delta)^{-1}\rho(\gamma) = g\left(\frac{\Phi_\gamma}{\Phi_\delta}, \frac{\xi_\gamma - \xi_\delta}{\Phi_\delta}\right). \tag{96}$$

In matrices, the same computation is

$$\begin{pmatrix} \Phi_\delta^{-1} & -\Phi_\delta^{-1}\xi_\delta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \Phi_\gamma & \xi_\gamma \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \Phi_\delta^{-1}\Phi_\gamma & \Phi_\delta^{-1}(\xi_\gamma - \xi_\delta) \\ 0 & 1 \end{pmatrix}.$$

If $\Phi_\gamma = \Phi_\delta = \Phi$, the relative element is a pure translation by $(\xi_\gamma - \xi_\delta)/\Phi$. The unnormalized endpoint difference is $\xi_\gamma - \xi_\delta$. These two scalar quantities differ by the common scale and should not be identified.

B.6 The elementary order square

Let

$$A_\mu = \begin{pmatrix} 1 & \mu \\ 0 & 1 \end{pmatrix}, \qquad S_q = \begin{pmatrix} q & 0 \\ 0 & 1 \end{pmatrix}.$$

Under the chronological convention, first adding and then scaling has matrix

$$S_q A_\mu = \begin{pmatrix} q & q\mu \\ 0 & 1 \end{pmatrix},$$

whereas first scaling and then adding has matrix

$$A_\mu S_q = \begin{pmatrix} q & \mu \\ 0 & 1 \end{pmatrix}.$$

Their endpoint difference is

$$(qx + q\mu) - (qx + \mu) = \mu(q-1).$$

Taking the second history as reference gives

$$(A_\mu S_q)^{-1}(S_q A_\mu) = \begin{pmatrix} 1 & \mu(1-q^{-1}) \\ 0 & 1 \end{pmatrix}.$$

Thus the relative group translation is $\mu(1-q^{-1})$, while the open endpoint defect is $\mu(q-1)$. The latter is the former multiplied by the common target scale q.

Closing the comparison into a group commutator produces yet another ordered quantity. For example,

$$S_q A_\mu S_q^{-1} A_\mu^{-1} = \begin{pmatrix} 1 & (q-1)\mu \\ 0 & 1 \end{pmatrix},$$

$$A_\mu S_q A_\mu^{-1} S_q^{-1} = \begin{pmatrix} 1 & (1-q)\mu \\ 0 & 1 \end{pmatrix}.$$

The signs differ because the loop orientations differ. These closed-loop calculations are recorded only to audit conventions; the torsion theory and its orientation choices are introduced in Section 8.

B.7 Exact perturbation propagation at every prefix

Let x and $\widetilde{x}$ be two initial values, and propagate both through the same history. Set

$$x_k = F_k(x), \qquad \widetilde{x}_k = F_k(\widetilde{x}).$$

From the prefix recursion,

$$\begin{aligned}\widetilde{x}_{k+1} - x_{k+1} &= (s_{k+1}\widetilde{x}_k + t_{k+1}) - (s_{k+1}x_k + t_{k+1}) \\ &= s_{k+1}(\widetilde{x}_k - x_k).\end{aligned}$$

Induction gives

$$\widetilde{x}_k - x_k = \left(\prod_{i=1}^{k} s_i\right)(\widetilde{x} - x) = \Phi_k(\widetilde{x} - x). \tag{97}$$

The translations cancel at each step; only the scale cocycle propagates the initial difference. Over $\mathbb{R}$, if $s_i = e^{\lambda_i}$, then

$$\frac{\widetilde{x}_k - x_k}{\widetilde{x} - x} = e^{\lambda_1 + \cdots + \lambda_k}.$$

No limiting process is involved in this identity.

B.8 Differential conversion check

Finally, on the positive real affine group let $\Phi = e^M$ and $\widehat{\xi} = e^{-M}\xi$. The product rule gives

$$\begin{aligned} d\widehat{\xi} &= d(e^{-M}\xi) \\ &= -e^{-M}\xi\, dM + e^{-M} d\xi \\ &= e^{-M}(d\xi - \xi dM).\end{aligned}$$

Therefore

$$d\xi - \xi dM = e^M d\widehat{\xi}.$$

This verifies that the translation component of $dg\, g^{-1}$ is the target scale times the differential of the source-normalized coordinate. It does not identify $dg\, g^{-1}$ with $g^{-1}dg$, whose translation component is $e^{-M}d\xi$.

C Hyperbolic-model calculations

This appendix records the coordinate checks used for the basic hyperbolic arithmetic expression space. Throughout, $\mu, \lambda > 0$,

$$G = \{(x, y) : x \in \mathbb{R},\ y > 0\}, \qquad (x, y)(x', y') = (x + yx', yy'),$$

and

$$g_{\mu,\lambda} = \frac{1}{y^2}\left(\frac{dx^2}{\mu^2} + \frac{dy^2}{\lambda^2}\right), \qquad a = -\frac{x}{y}.$$

C.1 Group coordinates and the invariant frame

The identity of G is $(0,1)$, and

$$(x,y)^{-1} = \left(-\frac{x}{y}, \frac{1}{y}\right).$$

For $g_0 = (x_0, y_0)$, left translation and its differential are

$$L_{g_0}(x,y) = (x_0 + y_0 x, y_0 y), \qquad dL_{g_0}(\partial_x) = y_0\partial_x, \qquad dL_{g_0}(\partial_y) = y_0\partial_y.$$

It follows that $y\partial_x$ and $y\partial_y$ are left-invariant. The normalized arithmetic frame

$$X_u = -\mu y\partial_x, \qquad X_v = -\lambda y\partial_y$$

is therefore left-invariant, and its dual coframe is

$$-\frac{dx}{\mu y}, \qquad -\frac{dy}{\lambda y}.$$

The sum of their squares is $g_{\mu,\lambda}$. The bracket is

$$\begin{aligned}[X_u, X_v] &= [-\mu y\partial_x, -\lambda y\partial_y] \\ &= -\mu\lambda y\partial_x = \lambda X_u,\end{aligned}$$

and hence $[X_u, X_v]a = \mu\lambda$, as required by the general arithmetic-frame identity.

The same invariance can be checked directly:

$$\begin{aligned}L^*_{(x_0,y_0)} g_{\mu,\lambda} &= \frac{1}{(y_0 y)^2}\left(\frac{(y_0 dx)^2}{\mu^2} + \frac{(y_0 dy)^2}{\lambda^2}\right) \\ &= g_{\mu,\lambda}.\end{aligned}$$

For a horocyclic chart, set

$$u = x, \qquad y = e^{\lambda v}.$$

Then

$$(u,v)(u',v') = (u + e^{\lambda v}u', v + v'),$$

and

$$X_u = -\mu e^{\lambda v}\partial_u, \qquad X_v = -\partial_v,$$

while

$$g_{\mu,\lambda} = e^{-2\lambda v}\frac{du^2}{\mu^2} + dv^2, \qquad a = -ue^{-\lambda v}.$$

The letters u, v in this chart are genuine coordinates, but the arithmetic frame is not the coordinate frame.

C.2 Inverse metric, gradient, and the eikonal identity

In the (x,y)-chart,

$$(g_{ij}) = \begin{pmatrix} \frac{1}{\mu^2 y^2} & 0 \\ 0 & \frac{1}{\lambda^2 y^2} \end{pmatrix}, \qquad (g^{ij}) = \begin{pmatrix} \mu^2 y^2 & 0 \\ 0 & \lambda^2 y^2 \end{pmatrix},$$

and

$$\det(g_{ij}) = \frac{1}{\mu^2\lambda^2 y^4}, \qquad d\,\mathrm{vol}_g = \frac{dx\wedge dy}{\mu\lambda y^2}.$$

Since

$$a_x = -\frac{1}{y}, \qquad a_y = \frac{x}{y^2} = -\frac{a}{y},$$

the gradient is

$$\begin{aligned}\nabla a &= g^{xx}a_x\partial_x + g^{yy}a_y\partial_y \\ &= -\mu^2 y\,\partial_x + \lambda^2 x\,\partial_y.\end{aligned}$$

Its squared norm is

$$\begin{aligned}|\nabla a|_g^2 &= g^{xx}a_x^2 + g^{yy}a_y^2 \\ &= \mu^2 + \lambda^2\frac{x^2}{y^2} \\ &= \mu^2 + \lambda^2 a^2.\end{aligned}$$

Alternatively,

$$X_u a = (-\mu y)\left(-\frac{1}{y}\right) = \mu, \qquad X_v a = (-\lambda y)\left(\frac{x}{y^2}\right) = \lambda a,$$

so the framed and eikonal verifications agree.

C.3 Curvature and completeness

Two calculations give the curvature normalization. First set

$$X = \frac{\lambda}{\mu}x.$$

This is a global diffeomorphism of upper half-planes, and

$$g_{\mu,\lambda} = \frac{1}{\lambda^2}\frac{dX^2+dy^2}{y^2}.$$

The standard metric $(dX^2+dy^2)/y^2$ is complete with curvature -1. Constant rescaling by λ^{-2} gives a complete metric of curvature $-\lambda^2$.

For a direct Cartan calculation, use the positively oriented orthonormal coframe

$$\omega^1 = \frac{dx}{\mu y}, \qquad \omega^2 = \frac{dy}{\lambda y}.$$

Then

$$d\omega^1 = \lambda\,\omega^1\wedge\omega^2, \qquad d\omega^2 = 0.$$

The Levi–Civita connection form ${\omega^1}_2 = -\lambda\omega^1$ satisfies

$$d\omega^1 = -{\omega^1}_2\wedge\omega^2, \qquad d\omega^2 = {\omega^1}_2\wedge\omega^1.$$

Its curvature form is

$$d{\omega^1}_2 = -\lambda\,d\omega^1 = -\lambda^2\,\omega^1\wedge\omega^2.$$

Thus $K = -\lambda^2$, consistently with the coordinate rescaling.

C.4 The Laplace–Beltrami operator

With the divergence-of-gradient convention,

$$\Delta_g f = \frac{1}{\sqrt{\det g}}\,\partial_i\Big(\sqrt{\det g}\,g^{ij}\partial_j f\Big)\,.$$

Here

$$\sqrt{\det g} = \frac{1}{\mu\lambda y^2}, \qquad \sqrt{\det g}\,g^{xx} = \frac{\mu}{\lambda}, \qquad \sqrt{\det g}\,g^{yy} = \frac{\lambda}{\mu}.$$

The last two coefficients are constant, so

$$\Delta_{g_{\mu,\lambda}} f = \mu^2 y^2 f_{xx} + \lambda^2 y^2 f_{yy}. \tag{98}$$

For $a = -x/y$,

$$a_{xx} = 0, \qquad a_{yy} = -\frac{2x}{y^3} = \frac{2a}{y^2}.$$

Equation (98) therefore gives

$$\Delta_{g_{\mu,\lambda}} a = 2\lambda^2 a.$$

C.5 Grid maps and their metric pullbacks

For

$$\mathsf{X}_s(x,y) = (x - sy, y), \qquad \mathsf{Y}_k(x,y) = \left(x, \frac{y}{k}\right),$$

one computes

$$(a \circ \mathsf{X}_s)(x,y) = -\frac{x - sy}{y} = a(x,y) + s$$

and

$$(a \circ \mathsf{Y}_k)(x,y) = -\frac{x}{y/k} = k\,a(x,y).$$

On differentials,

$$\mathsf{X}_s^* dx = dx - s\,dy, \qquad \mathsf{X}_s^* dy = dy,$$

and the transformed y-coordinate is still y. Hence

$$\mathsf{X}_s^* g_{\mu,\lambda} = \frac{1}{y^2}\left(\frac{(dx - s\,dy)^2}{\mu^2} + \frac{dy^2}{\lambda^2}\right).$$

For Y_k, the transformed coordinate and differentials are

$$y' = \frac{y}{k}, \qquad dx' = dx, \qquad dy' = \frac{dy}{k},$$

so

$$\mathsf{Y}_k^* g_{\mu,\lambda} = \frac{1}{y^2}\left(\frac{k^2 dx^2}{\mu^2} + \frac{dy^2}{\lambda^2}\right).$$

The cross term in the first formula and the unequal horizontal coefficient in the second show that these grid maps are not general isometries.

C.6 The Baumslag–Solitar composition check

For a positive integer n,

$$\mathsf{X}_s^n(x,y) = (x - nsy, y).$$

Using the standard right-to-left convention for composition,

$$\begin{aligned}(\mathsf{Y}_n^{-1} \circ \mathsf{X}_s^n \circ \mathsf{Y}_n)(x,y) &= (\mathsf{Y}_n^{-1} \circ \mathsf{X}_s^n)\left(x, \frac{y}{n}\right) \\ &= \mathsf{Y}_n^{-1}(x - sy, y/n) \\ &= (x - sy, y) \\ &= \mathsf{X}_s(x,y).\end{aligned}$$

Equivalently,

$$\mathsf{Y}_n \circ \mathsf{X}_s \circ \mathsf{Y}_n^{-1} = \mathsf{X}_s^n.$$

These are identities in the diffeomorphism group of $\mathbb{H}^2$. The metric pullback calculation above is why they are stated as grid relations rather than as relations in the hyperbolic isometry group.

D ACS and contact calculations

This appendix records sign checks and coordinate calculations used in Sections 8 and 9. The chronological convention is unchanged: entries in a history are performed from left to right. The ACS is oriented by $dA \wedge dM$, and the contact state space is oriented by $du \wedge da \wedge dv$.

D.1 The oriented ACS rectangle

Let

$$\gamma = (\mathsf{A}_p, \mathsf{M}_q), \qquad \delta = (\mathsf{M}_q, \mathsf{A}_p).$$

Their common terminal multiplicative charge is $M_* = q$. The chain $C_\gamma - C_\delta$ traverses the bottom edge from $(0,0)$ to $(p,0)$, the right edge to (p,q), the top edge back to $(0,q)$, and the left edge back to the origin. For $p, q > 0$ this is the positive boundary convention for $dA \wedge dM$. Since vertical edges do not contribute to $\eta_* = e^{q-M} dA$,

$$\begin{aligned}\int_{C_\gamma - C_\delta} \eta_* &= \int_0^p e^q \, dA + \int_p^0 1 \, dA \\ &= p(e^q - 1).\end{aligned}$$

On the other hand,

$$\int_{[0,p]\times[0,q]} e^{q-M} \, dA \wedge dM = p \int_0^q e^{q-M} \, dM = p(e^q - 1).$$

This verifies both the sign and the target-frame weight.

The same formula holds for signed p and q. More precisely, regard the rectangle as the oriented singular chain parametrized by

$$R_{p,q}(s,t) = (sp, tq), \qquad 0 \le s, t \le 1.$$

Its pulled-back orientation form is $pq \, ds \wedge dt$, so orientation reverses automatically when exactly one side is negative. Direct integration still gives

$$\int_{R_{p,q}} e^{q-M} \, dA \wedge dM = p(e^q - 1).$$

Thus no separate "unsigned area" convention is needed.

D.2 A closed charge path with nonzero evaluation defect

Consider the chronological loop

$$\ell = (\mathsf{A}_p, \mathsf{M}_q, \mathsf{A}_{-p}, \mathsf{M}_{-q}).$$

Its total ACS charge is $(0,0)$, the same as the empty history $\varnothing$, but

$$\begin{aligned}\nu_x(\ell) &= e^{-q}\big(e^q(x+p) - p\big)\\ &= x + p(1-e^{-q}).\end{aligned}$$

Hence

$$\tau(\ell, \varnothing) = p(1-e^{-q}).$$

Here $M_* = 0$, so $\eta_* = e^{-M}dA$. The ACS path of ℓ is the oriented rectangle boundary itself, and

$$\int_{C_\ell} e^{-M}\, dA = p - pe^{-q} = p(1-e^{-q}).$$

This example simultaneously checks the closed-loop sign and shows why zero total charge is weaker than operator neutrality.

For a self-intersecting charge path, the same computation is performed on a singular two-chain with integer multiplicities. Decomposing a filling into oriented rectangles adds their signed weighted integrals; overlap is counted with the sum of multiplicities. This is the precise replacement for speaking of "the region enclosed" by a self-intersecting loop.

D.3 Finite contact compositions

Write $p = \mu h$ and $q = \lambda k$. On the assignment coordinate, the two horizontal flows act by

$$A_h(a) = a + p, \qquad M_k(a) = e^q a.$$

The open comparison is

$$\begin{aligned}(M_k \circ A_h)(a) &= e^q(a+p),\\ (A_h \circ M_k)(a) &= e^q a + p,\end{aligned}$$

and therefore

$$(M_k \circ A_h)(a) - (A_h \circ M_k)(a) = p(e^q - 1).$$

For the closed chronological word A_h, M_k, A_{-h}, M_{-k}, the induced composition is

$$M_{-k} \circ A_{-h} \circ M_k \circ A_h,$$

and

$$\begin{aligned}(M_{-k} \circ A_{-h} \circ M_k \circ A_h)(a) &= e^{-q}\big(e^q(a+p) - p\big)\\ &= a + p(1-e^{-q}).\end{aligned}$$

Consequently

$$T_{\text{closed}} = p(1-e^{-q}) = e^{-q}p(e^q - 1) = e^{-q}T_{\text{open}}.$$

The equality after multiplication by e^q is a change to the common target frame, not an equality of the two raw finite defects.

D.4 Darboux coordinates and the Reeb field

Assume $\mu\lambda \neq 0$ and introduce natural coordinates

$$\widetilde{u} = \mu u, \qquad \widetilde{v} = \lambda v.$$

Then

$$\alpha = da - d\widetilde{u} - a\, d\widetilde{v}.$$

Set

$$x = \widetilde{v}, \qquad y = a, \qquad z = a - \widetilde{u}.$$

This is a global affine change of coordinates on $\mathbb{R}^3$, and

$$dz - y\, dx = da - d\widetilde{u} - a\, d\widetilde{v} = \alpha.$$

Thus the contact form is in the standard Darboux form. The arithmetic structure is not a new local contact-isomorphism class; it is the distinguished frame (D_u, D_v) and its interpretation by addition and scaling.

For completeness, the Reeb field R is determined by

$$\alpha(R) = 1, \qquad \iota_R d\alpha = 0.$$

Since $d\alpha = -\lambda da \wedge dv$, its kernel is spanned by ∂_u. Hence

$$R = -\frac{1}{\mu}\partial_u.$$

In the Darboux coordinates above this is $R = \partial_z$, as expected.

D.5 Expanded scalar-curvature calculation

Let $F = F(u, v, a)$. Expanding the two ordered derivatives gives

$$\begin{aligned} D_u D_v F &= F_{uv} + \lambda a F_{ua} + \mu F_{av} + \mu\lambda F_a + \mu\lambda a F_{aa}, \\ D_v D_u F &= F_{vu} + \mu F_{va} + \lambda a F_{au} + \mu\lambda a F_{aa}. \end{aligned}$$

Equality of the mixed coordinate derivatives leaves

$$D_u D_v F - D_v D_u F = \mu\lambda F_a.$$

Multiplying by $du \wedge dv$ yields

$$\mathcal{R}_H(F) = \mu\lambda(\partial_a F)\, du \wedge dv.$$

This is an antisymmetrized calculation on scalar fields. It neither defines δ_H on one-forms nor claims that a graded differential has been constructed.

E Equality and elementary neutrality examples

The examples below separate several equality tests without developing a general relation or loop theory. In each case, histories are marked chronological words, operators are affine maps, and endpoint values are obtained only after choosing an input.

E.1 Same value, different operators

Let

$$\gamma = (\mathsf{A}_1), \qquad \delta = (\mathsf{M}_{\log 2}).$$

At the input $x = 1$, both histories have endpoint value 2. Their induced operators are nevertheless

$$\nu_x(\gamma) = x + 1, \qquad \nu_x(\delta) = 2x,$$

which are unequal. Equality at one input therefore does not imply operator equality.

E.2 Same operator, different histories

The marked histories

$$\gamma = (\mathsf{A}_1, \mathsf{A}_2), \qquad \delta = (\mathsf{A}_2, \mathsf{A}_1)$$

are different chronological words, but both induce $x \mapsto x + 3$. They also have the same total charge. Thus operator equality does not recover a marked history, even when charge data are retained.

E.3 Same charge, nonzero relative torsion

For $p, q \neq 0$, the histories

$$\gamma = (\mathsf{A}_p, \mathsf{M}_q), \qquad \delta = (\mathsf{M}_q, \mathsf{A}_p)$$

have the same ACS endpoint (p, q). Their operators differ by the translation

$$\tau(\gamma, \delta) = p(e^q - 1).$$

The ACS endpoint is therefore insensitive to order, whereas evaluation is not. The weighted filling, rather than the terminal charge, detects the difference.

E.4 Zero endpoint defect, nontrivial history

For $p \neq 0$, the nonempty history

$$\ell = (\mathsf{A}_p, \mathsf{A}_{-p})$$

induces the identity operator and hence has zero endpoint defect for every input. It is nevertheless not the empty marked history. In particular, operator neutrality does not imply literal or historical equality.

A slightly different phenomenon occurs for

$$(\mathsf{A}_p, \mathsf{M}_q, \mathsf{A}_{-p}, \mathsf{M}_{-q}).$$

This history has zero total ACS charge but induces translation by $p(1 - e^{-q})$. Thus charge neutrality does not imply operator neutrality.

E.5 Open torsion and closed-loop drift

For additive side $p = \mu h$ and logarithmic multiplicative side $q = \lambda k$, the two open orders differ by

$$T_{\text{open}} = p(e^q - 1),$$

whereas the chronological closed loop has vertical drift

$$T_{\text{closed}} = p(1 - e^{-q}) = e^{-q} T_{\text{open}}.$$

They have the common first-order density $\mu\lambda$, but they are distinct finite quantities. Consequently local order torsion, raw closed-loop holonomy, and infinitesimal curvature should not be used as interchangeable notions without the normalization or limiting operation that relates them.

These examples exhibit the hierarchy needed in the main text:

$$\text{marked history} \longrightarrow \text{operator} \longrightarrow \text{endpoint value},$$

with total charge providing an additional, order-forgetting projection. No converse implication between adjacent levels holds in general.

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